

**SUSTAINABLE SELF HEALING CEMENTITIOUS
MATERIAL: BIO-MINERALIZATION AND AI-DRIVEN
SOLUTIONS FOR CLIMATE ADAPTATION**

Submitted by

SHAYAN AHMED AZAD (22PWCIV5977)

SYED HASNAT SHAH (22PWCIV5897)

MUHAMMAD SHAHZAD (22PWCIV5799)

MOHAMMAD USMAN JALAL (22PWCIV5909)

Submitted for the partial fulfillment of the Degree of Bachelor of Science in
Civil Engineering.

Supervised by:

Prof. Dr. Yasir Irfan Badrashi

Internal Examiner:

Prof. Dr. Khan Shahzada

External Examiner:

Dr. Zain Ul Abidin

SUSTAINABLE SELF HEALING CEMENTITIOUS MATERIAL: BIO-MINERALIZATION AND AI-DRIVEN SOLUTIONS FOR CLIMATE ADAPTATION



by

SHAYAN AHMED AZAD

SYED HASNAT SHAH

MUHAMMAD SHAHZAD

MOHAMMAD USMAN JALAL

A thesis

Presented to the University of Engineering and Technology, Peshawar

In partial fulfillment for the degree requirement of

Bachelor of Science

In

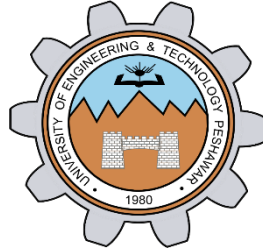
Civil Engineering

2025

Peshawar, Khyber Pakhtunkhwa, Pakistan

UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Civil Engineering
University of Engineering and
Technology, Peshawar



SUSTAINABLE SELF HEALING CEMENTITIOUS MATERIAL: BIO-MINERALIZATION AND AI-DRIVEN SOLUTIONS FOR CLIMATE ADAPTATION

Group Number: 39

Batch: 2022 – 2026

Group Members

Names	Registration No
Shayan Ahmed Azad	22PWCIV5977
Syed Hasnat Shah	22PWCIV5897
Muhammad Shahzad	22PWCIV5799
Mohammad Usman Jalal	22PWCIV5909

Checked and verified by

Approved by

Dr. Sikandar Hayat Sajid
CED UET PESHAWAR

Prof. Dr. Amjad Naseer
Chairman, CED UET PESHAWAR

University of Engineering & Technology, Peshawar. All Rights Reserved - JULY 2026

AUTHOR'S DECLARATION

We hereby declare that we are the sole authors of this thesis. This is a true copy of the thesis, including any required final revisions, as accepted by our examiners. It is further declared, that we have fulfilled all the requirements in line with the Quality Assurance guidelines of the Higher Education Commission (HEC).

We understand that our thesis may be made electronically available to the public.

Signature and Date

Shayan Ahmed Azad
22PWCIV5977
22pwciv5977@uetpeshawar.edu.pk

Syed Hasnat Shah
22PWCIV5897
22pwciv5897@uetpeshawar.edu.pk

Muhammad Shahzad
22PWCIV5799
22pwciv5799@uetpeshawar.edu.pk

Mohammad Usman Jalal
22PWCIV5909
22pwciv5909@uetpeshawar.edu.pk

Abstract

Cracking reduces the durability of cementitious materials by facilitating the ingress of water and aggressive agents; therefore, bacterial self-healing offers a potential means of limiting crack-related deterioration. This study evaluated the effects of calcium lactate dosage and healing temperature on the surface crack closure and compressive-strength recovery of *Bacillus subtilis* mortar and developed machine-learning models for performance prediction. Mortar cubes were prepared at a cement–sand ratio of 1:2 and a water–cement ratio of 0.45 using a bacterial suspension at 10^8 cells/mL. A plain control mix and three bacterial batches (A, B, and C) containing 10%, 0.5%, and 1.5% calcium lactate by cement mass, respectively, were examined. After 28 days of water curing, pre-crack compressive strength was determined, controlled surface cracks were induced, and Batches B and C were subjected to 28-day healing at -15 , 20 , 40 , and 50 °C. One representative crack-width sequence per batch–temperature condition was measured using ImageJ on Days 0, 7, 14, 21, and 28, followed by post-healing compressive-strength testing. Batch C produced the highest pre-crack mean strength of 32.18 MPa, whereas Batch A produced the lowest mean of 12.35 MPa. Complete apparent surface closure was recorded by Day 28 at 20 , 40 , and 50 °C, while the -15 °C specimens achieved approximately 50% closure. Batch C containing 1.5% calcium lactate at 40 °C was the best-performing tested condition, reaching complete closure by Day 21 and attaining a post-healing mean strength of 27.28 MPa and 84.8% strength recovery. Two XGBoost models used temperature, calcium lactate dosage, and Day-0 crack width to predict Day-28 healing efficiency and post-healing compressive strength, achieving cross-validated mean absolute errors of 1.00 percentage point and 2.18 MPa, respectively. The results indicated that calcium lactate dosage and temperature influenced the healing response, while the machine-learning component provided a proof of concept using eight independent conditions and therefore required further validation.

Acknowledgements

The completion of various phases of research study has been a total blessing of Almighty Allah due to which the overall research was made possible.

We would like to express our heartfelt gratitude to our respected supervisor, **Prof. Dr. Yasir Irfan Badrashi**, for his continuous guidance, encouragement, and insightful feedback throughout the course of this research. His valuable input played a crucial role in shaping this study.

We are also sincerely thankful to **Prof. Dr. Syed Muhammad Ali** and **Dr. Adeel Arshad** for their academic support and helpful suggestions that enriched the quality of our work.

We express our sincere gratitude to **Dr. Hina Fazal**, Senior Scientific Officer at the Pakistan Council of Scientific and Industrial Research (PCSIR) Peshawar, for her generous cooperation and for facilitating the provision of the bacterial culture required for this research. Her support was invaluable to the successful completion of the experimental work.

We are also deeply grateful to **Prof. Dr. Waqas Ahmed**, Department of Biotechnology and Genetic Engineering, The University of Agriculture, Peshawar, for his kind cooperation in arranging the incubators used during the experimental phase of this study. His support and assistance are sincerely appreciated.

Table of Contents

Contents

CHAPTER 1	1
INTRODUCTION	1
1.1 Background of the Study	1
1.3 Role of calcium lactate in Bacterial Self-Healing.....	3
1.5 Sustainability and Climate Adaptation Benefits	6
1.6 Artificial Intelligence in Self-Healing Concrete	7
1.6.1 Completed Machine-Learning Component	9
1.7 Research Gap	9
1.8 Problem Statement	10
1.9 Research Aim	11
1.10 Research Objectives.....	11
1.11 Research Questions	12
1.12 Scope of Study	12
1.13 Significance of the Study	13
CHAPTER 2	14
LITERATURE REVIEW	14
2.1 Introduction.....	14
2.2 Cracking and Durability Problems in Cementitious Materials	15
2.3 Concept of Self-Healing Concrete	16
2.4 Biomineralization and MICP Mechanism.....	17
2.5 Bacteria Used in Self-Healing Concrete	18
2.6 <i>Bacillus subtilis</i> in Self-Healing Cementitious Materials	19
2.7 Role of calcium lactate in Bacterial Self-Healing.....	20

2.8 Nutrient Optimization and calcium lactate Concentration.....	22
2.9 Bacterial Incorporation and Protection Methods	23
2.10 Effect of Temperature on Biomineralization and Self-Healing	24
2.11 Crack-Healing Efficiency Evaluation Methods	25
2.12 Compressive Strength Recovery in Bacterial Concrete.....	26
2.13 Durability Improvement through Bacterial Self-Healing	27
2.14 Sustainability and Life-Cycle Benefits	28
2.15 Artificial Intelligence and Machine Learning in Self-Healing Concrete	29
2.16 Integration of AI with Bacterial Self-Healing Concrete	31
2.16.1 Relevance of XGBoost to the Present Study.....	32
2.17 Critical Review of Previous Studies	33
2.18 Summary of Literature Review	34
CHAPTER 3 METHODOLOGY	35
3.1 Introduction.....	35
3.2 Research Design and Experimental Program	35
3.2.1 Experimental Groups	35
3.2.2 Experimental Variables.....	36
3.2.3 Specimen Allocation	36
3.3 Materials	37
3.3.1 Cement.....	37
3.3.2 Fine Aggregate	37
3.3.3 Mixing and Curing Water	37
3.3.4 Calcium Lactate	37
3.3.5 Bacterial Culture and Nutrient Medium	38
3.4 Preparation of the bacterial suspension.....	38
3.4.1 Sterilization of the Medium and Glassware	38
3.4.2 Aseptic Transfer of the Parent Culture	38

3.4.3 Incubation and Concentration Adjustment	38
3.5 Mortar Mix Design and Specimen Preparation	40
3.5.1 Base Mix Proportions	40
3.5.2 Manual Mixing and Casting	41
3.5.3 Demolding and Water Curing	41
3.6 Pre-crack Compressive-Strength Test	44
3.6.1 Test Specimens and Preparation	44
3.6.2 Loading Procedure	44
3.6.3 Compressive-Strength Calculation	44
3.7 Controlled Crack Induction.....	45
3.7.1 Crack-Induction Arrangement.....	45
3.7.2 Loading and Stopping Criterion	45
3.7.3 Day 0 Documentation	46
3.8 Temperature-Controlled Healing	46
3.8.1 Selection of Batches	46
3.8.2 Healing Environments	46
3.9 Crack Monitoring and Healing Evaluation	47
3.9.1 Image-Acquisition Schedule	47
3.9.2 ImageJ Measurement Procedure	47
3.9.3 Crack-Healing Efficiency	47
3.10 Post-Healing Compressive-Strength Test	48
3.10.1 Specimen Preparation and Loading	48
3.10.2 Strength-Recovery Calculation	48
3.11 Data Collection and Analysis	49
3.11.1 Recorded Data	49
3.11.2 Descriptive Analysis	49
3.11.3 Machine-Learning Dataset Construction	49
3.11.4 Model Inputs and Prediction Targets	49

3.11.5 XGBoost Model Configuration and Training	50
3.11.6 Cross-Validation and Performance Measures	51
3.11.7 Final Model Fitting, Saving, and Prediction Domain.....	51
CHAPTER 4	53
RESULTS AND DISCUSSION.....	53
4.1 Introduction.....	53
4.2 Pre-Crack Compressive Strength.....	53
4.2.1 Results.....	53
4.2.2 Discussion	55
4.3 Crack Induction and Selection of Specimens for the Healing Study	55
4.3.1 Crack-Induction Observations	55
4.3.2 Rationale for Excluding Batch A.....	56
4.4 Crack-Healing Performance of Batches B and C	57
4.4.1 Crack-Width Reduction Over Time	57
4.4.2 Photographic Evidence of Healing	58
4.4.3 Discussion	63
4.5 Post-Healing Compressive Strength	64
4.5.1 Results.....	64
4.5.2 Discussion	66
4.6 Healing Efficiency	67
4.6.1 Calculation and Results	67
4.6.2 Best-Performing Combination.....	68
4.7 Comparative Analysis	69
4.7.1 Effect of calcium lactate Concentration	69
4.7.2 Effect of Temperature	69
4.8 Comparison with Previous Studies	69
4.9 Machine-Learning Prediction	70

4.9.1 Completed Model Configuration and Validation	70
4.9.2 Prediction of Day-28 Healing Efficiency	71
4.9.3 Prediction of Post-Healing Compressive Strength.....	71
4.9.4 Condition-Level Predictions and Feature Importance	72
4.9.5 Practical Interpretation and Limitations	72
4.10 Summary of Key Findings	73
CHAPTER 5	75
RECOMMENDATIONS	75
5.1 Recommendations for Experimental Application.....	75
5.2 Recommendations for Further Research.....	76
REFERENCE LIST	77

List of Figures

FIGURE 3- 2 STERILIZED LIQUID NUTRIENT BROTH PREPARED FOR BACTERIAL INOCULATION.	39
FIGURE 3- 1 PARENT CULTURE OF <i>BACILLUS SUBTILIS</i> ON A NUTRIENT AGAR PLATE.	39
FIGURE 3- 3 AUTOCLAVE USED FOR STERILIZATION OF NUTRIENT MEDIUM AND LABORATORY GLASSWARE.....	39
FIGURE 3- 4 SHAKING INCUBATOR USED FOR PREPARATION OF THE BACTERIAL SUSPENSION.	39
FIGURE 3- 5 ASEPTIC TRANSFER OF A <i>BACILLUS SUBTILIS</i> COLONY INTO NUTRIENT BROTH.	39
FIGURE 3- 6 FLOWCHART FOR PREPARATION OF THE <i>BACILLUS SUBTILIS</i> SUSPENSION.	40
FIGURE 3- 7 CASTING, DEMOLDING, AND WATER-CURING SEQUENCE FOR MORTAR CUBES.	42
FIGURE 3- 8 SIEVING OF SAND.....	43
FIGURE 3- 9 ADDITION OF CALCIUM LACTATE.	43
FIGURE 3- 10 : BACTERIAL SUSPENSION	43
FIGURE 3- 11: BACTERIAL SPECIMENS.....	43
FIGURE 3- 12 : 1 MM WIRE PLACED UNDER THE SPECIMEN.....	45
FIGURE 3- 13 :CONTROLLED CRACK INDUCTION.	45
FIGURE 3- 14 SNAPSHOT OF THE COMPLETED XGBOOST MODEL-DEVELOPMENT AND	52
FIGURE 4 - 1 MEAN PRE-CRACK COMPRESSIVE STRENGTH WITH SAMPLE STANDARD-DEVIATION ERROR BARS.....	54
FIGURE 4 - 2 PHOTOGRAPHIC COMPARISON OF THE BATCH A CRACK AT DAY 0 AND APPROXIMATELY 24 H AFTER INDUCTION.	56
FIGURE 4 - 3 REPRESENTATIVE CRACK-WIDTH REDUCTION FOR BATCHES B AND C AT THE FOUR HEALING TEMPERATURES.	58
FIGURE 4 - 4 REPRESENTATIVE IMAGEJ CRACK-MONITORING SEQUENCES FOR BATCH B (0.5% CALCIUM LACTATE) AND BATCH C (1.5% CALCIUM LACTATE) AT -15°C FROM DAY 0 TO DAY 28.....	59
FIGURE 4 - 5 REPRESENTATIVE IMAGEJ CRACK-MONITORING SEQUENCES FOR BATCH B (0.5% CALCIUM LACTATE) AND BATCH C (1.5% CALCIUM LACTATE) AT 20°C FROM DAY 0 TO DAY 28.....	60

FIGURE 4 - 6 REPRESENTATIVE IMAGEJ CRACK-MONITORING SEQUENCES FOR BATCH B (0.5% CALCIUM LACTATE) AND BATCH C (1.5% CALCIUM LACTATE) AT 40 °C FROM DAY 0 TO DAY 28.....	61
FIGURE 4 - 7 REPRESENTATIVE IMAGEJ CRACK-MONITORING SEQUENCES FOR BATCH B (0.5% CALCIUM LACTATE) AND BATCH C (1.5% CALCIUM LACTATE) AT 50 °C FROM DAY 0 TO DAY 28.....	62
FIGURE 4 - 8 MEAN POST-HEALING COMPRESSIVE STRENGTH WITH SAMPLE STANDARD-DEVIATION ERROR BARS.	65
FIGURE 4 - 9 STRENGTH RECOVERY RELATIVE TO THE CORRESPONDING PRE-CRACK COMPRESSIVE- STRENGTH MEAN.	66
FIGURE 4 - 10 HEALING-EFFICIENCY DEVELOPMENT FOR THE REPRESENTATIVE SPECIMENS AT EACH TEMPERATURE.	68
FIGURE 4 - 11 SNAPSHOTS OF THE COMPLETED MACHINE-LEARNING PREDICTION INTERFACE.	73

List of tables

TABLE-3- 1 CONTROL AND EXPERIMENTAL MORTAR GROUPS	36
TABLE-3- 2 SPECIMEN ALLOCATION AND TESTING STAGES	37
TABLE-3- 3 BASE MORTAR PROPORTIONS AND SPECIMEN CHARACTERISTICS	41
TABLE-3- 4 TEMPERATURE CONDITIONS USED DURING THE 28-DAY HEALING PERIOD	46
TABLE-3- 5 CRACK-MONITORING SCHEDULE	47
TABLE-3- 6 INPUT AND TARGET VARIABLES USED IN THE COMPLETED MACHINE-LEARNING MODELS	50
TABLE-3- 7 XGBOOST CONFIGURATION AND VALIDATION STRATEGY	51
TABLE - 4 - 1 PRE-CRACK COMPRESSIVE STRENGTH AFTER 28-DAY CURING.....	54
TABLE - 4 - 2 REPRESENTATIVE CRACK-WIDTH MEASUREMENTS DURING THE 28-DAY HEALING PERIOD	57
TABLE - 4 - 3 POST-HEALING COMPRESSIVE STRENGTH AND STRENGTH RECOVERY AFTER 28 DAYS	64
TABLE - 4 - 4 : HEALING-EFFICIENCY SUMMARY FOR REPRESENTATIVE SPECIMENS.....	67
TABLE - 4 - 5 CONDITION-LEVEL MEASURED AND CROSS-VALIDATED PREDICTIONS	72

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Concrete is one of the most vital construction materials in the world. It is extensively applied in structures like bridges, dams, buildings, roads, ports, tunnels, and retaining walls, and in various infrastructure sectors. This is largely because of its strength when compressed, the raw material used, low cost, and its ease to cast in various shapes. Modern society has been able to create strong, large scale constructions by using cementitious materials. But concrete is problematic too, in terms of durability and the environment. Cement manufacturing is an important emitter of CO₂ and the global demand for cement is growing due to urbanization and infrastructure development [1]. Although concrete performs well in compression, it is weak in tension. This weakness can cause cracks to occur as a result of plastic shrinkage, drying shrinkage, thermal stresses, external loading, freeze–thaw action, chemical attack, settlement and corrosion of reinforcement [2]–[4]. Small cracks will not immediately greatly affect the load carrying capacity of concrete. These, however, provide conduits for water, oxygen, chlorides, sulphates, carbon dioxide, and other harmful agents to penetrate the concrete. These agents affect the cement matrix, and may also initiate corrosion of steel reinforcement. However, when corrosion sets in, the steel begins to expand, and the internal pressure rises, thus causing more cracks to form. This results in the degradation of durability, safety and service life of reinforced concrete structures [2, 5]. As the climate changes cracking becomes more serious. Climate change can lead to an increase in environmental temperature, alteration of rainfall and the concentration of CO₂, and increase in exposure to aggressive wetting and drying cycles of structures. These alterations can speed up the carbonation, corrosion, chloride ingress and other degradation mechanisms of the concrete structure [2]. The original design lives are now coming to an end, with infrastructure in many regions now facing challenges that were not fully anticipated. So the design and material related to durability has gained importance in civil engineering, along with material adaptation to climate. Traditional crack repair methods include epoxy injection, polymer grouting, cementitious repair mortar, surface

coating, and external sealing. These methods can be effective, but they are usually applied only after cracks become visible. They also require labour, inspection, access to the damaged zone, and additional repair materials. Some cracks are difficult to reach, especially in underground structures, marine structures, deep foundations, water-retaining structures, and heavily reinforced members. Repair activities also add cost and environmental burden because more materials, transport, and energy are required [4]–[6].

For these reasons, researchers have developed the concept of self-healing concrete. Self-healing concrete is a material that can repair small cracks with little or no external human intervention. Healing may occur through continued hydration, calcium carbonate formation, polymer release, crystalline admixtures, or biological activity. Among these methods, bacteria-based self-healing concrete has received strong research attention because it can produce calcium carbonate inside cracks through biomineralization [3], [5]–[8]. This approach is especially attractive because the healing product, calcium carbonate, is compatible with the cementitious matrix.

Self-healing concrete aims to address the detrimental impact of crack by self-healing the cracks once they form. In normal cementitious materials, a certain amount of autogenous healing can take place due to the partial partial hydration of unhydrated cement particles or the natural formation of calcium carbonate in cracks. The natural healing is typically only possible for very small cracks and requires high moisture content, crack width, and number of unhydrated cement [4] [5] [8]. Repeat or larger cracks require more healing agents. One of the most promising self-healing methods is biological self-healing. The bacteria are added to the concrete/mortar system along with an appropriate nutrient and calcium source in this approach. As the crack opens up, the water and oxygen infiltrate the crack and stimulate the bacteria. The bacteria then participate in biochemical reactions which result in calcium carbonate precipitation. The precipitated calcium carbonate plugs the crack, lowers the permeability of the crack, and partially restores the strength [6]–[10]). Biological crack healing is mainly due to the process called Microbially Induced Calcium Carbonate Precipitation (MICP). MICP is a biomineralization method where the microorganisms produce favorable chemical conditions for the formation of calcium carbonate. Calcium carbonate precipitation requires calcium ions, carbonate ions, pH,

dissolved inorganic carbon and nucleation sites. The bacterial metabolism can also promote carbonate formation [5, 7, 11] and the cell surface could serve as a nucleation site. The overall outcome in this case is the growth of crystals of CaCO_3 , which are usually calcite. Various species of bacteria have been investigated for self-healing concrete, including *Bacillus subtilis*, *Bacillus sphaericus*, *Bacillus megaterium*, *Bacillus pseudofirmus*, *Sporosarcina pasteurii* or other alkaliphilic or spore-forming bacteria [3, 5, 12]. The genus *Bacillus* is often more popular because many of the species of *Bacillus* are capable of forming spores. Spores are encapsulated, persisting form, which enable bacteria to withstand adverse environmental conditions, including drying, high levels of alkalization, nutrient deficiency and high temperatures. This is crucial in concrete, since the concrete pore solution is very alkaline, the pH is often higher than 12 during the early hydration period [5]-[6]-[8]-[9]. *Bacillus subtilis* is especially relevant to this research because it is a spore-forming, non-pathogenic bacterium that has been used in several self-healing cementitious studies. Previous research has shown that *Bacillus subtilis* can improve crack healing and mechanical performance of mortar and concrete through calcium carbonate precipitation [9], [10], [13]. In a recent mortar study, *Bacillus subtilis* improved self-healing performance under different crack widths and curing conditions, and microstructural observations confirmed mineral deposition in cracks [9]. Other studies using *Bacillus subtilis* with calcium lactate also reported crack repair, strength improvement, permeability reduction, and calcite formation [10], [14], [15].

The use of bacteria provides several advantages over conventional repair methods. First, the healing process can start from inside the crack when water enters. Second, the healing product is compatible with cementitious materials. Third, the method can reduce permeability and slow down the entry of aggressive substances. Fourth, it can reduce the need for repeated manual repairs. Finally, the method supports sustainable construction by extending service life and reducing repair material demand [3], [5], [6], [12].

1.3 Role of calcium lactate in Bacterial Self-Healing

Bacteria require a suitable nutrient source to remain active and to support biomineralization. Several nutrient and calcium sources have been used in bacterial concrete, such as calcium lactate, calcium acetate, calcium nitrate, calcium formate, urea,

yeast extract, and other organic compounds [5], [7], [16], [17]. Among these, calcium lactate is one of the most commonly studied nutrients in bacteria-based self-healing concrete.

Calcium lactate has two important roles. First, it provides calcium ions that are needed for calcium carbonate precipitation. Second, it acts as an organic nutrient that can be metabolized by bacteria. When bacteria consume calcium lactate under suitable conditions, carbonate species are produced, and these can react with calcium ions to form calcium carbonate. The product can fill cracks and pores and improve the density of the matrix [6], [16]–[18].

The use of calcium lactate in self-healing concrete was strongly developed in early bacteria-based healing studies. Wiktor and Jonkers quantified crack healing in bacteria-based concrete using bacterial spores and calcium lactate as part of the healing agent system [18]. Their study showed that bacterial concrete could heal wider cracks than control concrete. Jonkers and co-workers also showed that bacteria can be used as self-healing agents for sustainable concrete development [6]. These studies formed the foundation for later research using calcium lactate in microbial concrete.

Vijay and Murmu investigated the effect of calcium lactate on mechanical properties of Compressive strength and self healing of cracks in microbial concrete [16]. They found that calcium lactate could help heal wounds with microorganisms; however, the dosage must be regulated. If the nutrient is too low, it can cause insufficient precipitation of calcium carbonate; if the nutrient is too high, it can cause the cementitious matrix to be disrupted or organic matter to remain unreacted. Abo Sabah and co-workers also studied the optimization of calcium lactate content in bio-concrete and found that the compressive strength, tensile strength, flexural strength and durability indicators of bio-concrete were enhanced when the calcium lactate content was optimized [17]. There are some studies to support the significance of calcium lactate optimization in mortar and concrete. Muhammad et al. examined *Bacillus subtilis*, as a self-healing material, and investigated strength influence of curing time and calcium lactate content in the cement mortar [14]. The combined and individual effect of *Bacillus subtilis* and calcium lactate on cement mortar was studied by the same research group [15]. The results of their research have

significance because their data demonstrate that using bacteria and nutrient together is more effective than using either alone. In this work, Chaerun and co-workers also produced bacteria containing calcium lactate pentahydrate in the mortar, which resulted in better properties of the mortar and self-healing occurrence [19]. In this thesis, calcium lactate was used at 0.5%, 1.5%, and 10%. These doses have been chosen to investigate low, moderate and high nutrient levels. The 0.5% dosage is considered low nutrient condition, which can promote biomineralization but may not contain sufficient calcium lactate for good healing. The 1.5% dosage is a moderate nutrient content that could be sufficient to supply adequate calcium and metabolic support without causing significant disruption of the matrix. This amount of 10% dosage is a high nutrient level which may lead to increased available calcium but could also have a detrimental impact on performance if it becomes too high for the mortar system. In the current experimental study, the optimized dose as per the requirement of optimizing the nutrients in bacterial self healing systems was 1.5% calcium lactate which was best among the tested dosages. At normal room temperature, such as around 20 °C, bacterial activity and cementitious curing are usually more favourable than at freezing or near-freezing conditions. Many laboratory studies on bacterial concrete are conducted at room temperature because it provides stable conditions for curing, microbial activity, and strength development [5], [9], [16]. However, the optimum temperature may vary depending on bacterial species, nutrient type, curing conditions, and crack environment.

Moderate-high temperature can increase bacterial metabolic activity and calcium carbonate precipitation up to a certain limit. In many biological and chemical systems, increasing temperature increases reaction rate until the bacteria or enzymes become stressed. This explains why some MICP processes perform better at warmer conditions. In this thesis, the best performance was found at 40 °C. This result suggests that *Bacillus subtilis* remained active at this temperature and that the reaction conditions supported strong calcium carbonate formation and compressive-strength recovery.

However, very high temperature can reduce bacterial survival and activity. Wang and co-workers reported that 50 °C conditions could significantly reduce bacterial activity within a short period in MICP systems [21]. This is important because it explains why 50 °C may

not always produce the best healing, even though chemical reaction rates may be faster. At excessive temperature, bacterial cells may become damaged, enzymes may lose activity, and moisture may be reduced. These effects can limit biomineralization and reduce crack-healing efficiency.

Recent concrete studies also show the influence of temperature on microbial self-healing performance. Ahmad et al. evaluated the effectiveness of *Bacillus* strains at different temperatures such as room temperature, 0°C, and sub-zero temperatures using SEM, EDS, XRD, ultrasonic testing, compressive strength, and flexural strength [20]. They found that the bacterial self-healing was highly dependent on the strain and temperature. This is in accordance with the need to test the performance of *Bacillus subtilis* at different temperatures.

The present study used -15 °C, 20 °C, 40 °C, and 50 °C. These temperatures are the cold, normal, warm and high temperatures. These four conditions provide a rationale for understanding the response of *Bacillus subtilis* mortar in various climates. The finding of 40 °C as optimal is significant for climate adaptation since the warm service conditions are found in many regions. The diminished performance at extremely low or high temperatures, however, demonstrates that bacterial self-healing systems have to be tuned to the environment they are used in.

1.5 Sustainability and Climate Adaptation Benefits

Sustainability is a major reason for developing self-healing cementitious materials. Cement production contributes significantly to global carbon dioxide emissions [1]. In addition, repair and replacement of damaged concrete structures require more cement, aggregates, water, energy, transport, and labor. If a material can heal small cracks and slow down deterioration, it can reduce repair frequency and extend service life. This can reduce life-cycle environmental impact, especially in large infrastructure systems [5], [6], [12].

Bacterial self-healing concrete supports sustainability in several ways. First, it can seal cracks and reduce permeability. Lower permeability reduces the movement of water, oxygen, chlorides, and carbon dioxide into the concrete. This can slow down reinforcement corrosion and chemical deterioration [10], [11], [18]. Second, calcium carbonate

precipitation can fill pores and cracks, improving the density of the matrix and supporting strength recovery [9], [16], [19]. Third, autonomous healing can reduce the need for external repair materials and repeated site work [4], [5].

This study also includes climate adaptation. Infrastructure under conditions of temperature fluctuation, wetting and drying cycles, high carbon dioxide levels, and harsh environments. These conditions can lead to expansion of cracks and decrease in durability [2]. Self-healing mortar and concrete are a way to make infrastructure more resilient against climate change by filling in small cracks before they turn into sizeable durability issues. It is helpful in structures that are subjected to fluctuation of weather conditions such as bridges, pavements, coastal structures, retaining walls, water tanks, underground concrete members, etc.

Crack closure is not the only aspect that determines the sustainability value of bacterial self-healing concrete. It is also related to service-life extension. The longer the structure stands, the fewer there are of occasions when it needs to be demolished and replaced, and the fewer materials that need to be produced. Prolonged life results in reduced carbon footprint for infrastructure. Recent reviews have focused on the durability enhancement, reduced maintenance, reduced environmental impact, and practical resilience aspects of microbial self-healing concrete [5, 12, 22]. Hence, the aim of this research is aligned with the broader one of sustainable and climate-adapted construction.

1.6 Artificial Intelligence in Self-Healing Concrete

1.6 Artificial Intelligence in Self-Healing Concrete

In the field of civil engineering and construction materials research, Artificial Intelligence is playing a more and more significant role. There are several factors to consider when determining the behavior of concrete: the amount of cement used, the ratio of water to cement, the properties of the aggregates, the age of the concrete, the temperature, additives used to modify the properties of concrete, amount of bacteria in the concrete, nutrient dosage, and conditions of exposure. These are not linear relationships between the variables. Such complexity makes it difficult to accurately predict the compressive strength, crack healing, durability or long-term performance of traditional empirical equations. The use of AI and machine learning can aid in the identification of patterns

within the experimental data and predict the performance of a material with fewer trial mixes [23]–[28].

Machine learning methods such as Random Forest, Support Vector Regression, Artificial Neural Networks, Adaptive Neuro-Fuzzy Inference Systems, and Deep Learning have been used to predict concrete compressive strength and related properties [23], [25]–[28]. For bacterial concrete, Al mohammed and co-workers compared several soft computing methods for predicting compressive strength and found that machine learning models can effectively predict bacterial concrete strength using inputs such as cement, aggregate, sand, water–cement ratio, curing time, bacteria percentage, and sand type [23].

Vijay and Murmu applied Artificial Neural Networks to microbial concrete containing *Bacillus subtilis* and calcium lactate [24]. Their model used calcium lactate dosage, bacterial concentration, and specimen age as input variables, with compressive strength as the output. This is directly related to the present thesis because the current study also uses *Bacillus subtilis*, calcium lactate concentration, and compressive-strength recovery. Sri and co-workers also used ANN and ANFIS models to evaluate sustainable self-healing concrete with varying calcium lactate content [29]. These studies show that AI can support the optimization of bacterial concrete mix design.

There are multiple methods to implement AI in bacterial self-healing concrete. First, it has the potential to forecast the compressive strength before and after healing. Secondly, it is capable to estimate healing efficiency as function of bacterial concentration, nutrient dosage, crack width, curing time and temperature. Third, cracks can be measured in terms of their width and closure through image processing and deep learning for photographs and microscopy images. Fourth, AI can be used to forecast indicators of durability, like permeability, water absorption, chloride resistance, and carbonation risk. Fifth, Explainable AI techniques can provide insights into the importance of various variables affecting performance [25]–[28], [30].

The machine-learning component is not a new concept in this thesis, but a component of a completed study. To build two preliminary XGBoost regression models, the measured data of *Bacillus subtilis* mortar were used, which included the information of calcium lactate dosage, healing temperature and Day-0 crack width. The models offer an early-condition

assessment method for determining the healing efficiency after 28 days and post-healing compressive strength in the assessed domain but are not an optimum model nor do they supplant additional experimental validation.

1.6.1 Completed Machine-Learning Component

Earlier drafts described the predictive component as planned work. The component was subsequently completed as part of the present study. Two supervised XGBoost regression models were developed: one for predicting Day-28 healing efficiency and one for predicting post-healing compressive strength.

Both models use three inputs that are available immediately after controlled crack induction: healing temperature, calcium lactate concentration, and Day-0 crack width. The first model provides an early estimate of Day-28 surface healing without requiring Day-7, Day-14, Day-21, or Day-28 crack-width measurements. The second model provides a preliminary estimate of the post-healing compressive strength from the same initial-condition variables.

This model structure was selected to give the predictive component a practical purpose. Healing efficiency can be calculated exactly after Day 28 when both the initial and final crack widths are known; therefore, a model that uses Day-28 crack width to predict Day-28 healing efficiency would reproduce information already contained in the calculation. Restricting the inputs to initial conditions instead provides a forecast before the final healing observation and compressive-strength test.

The use of XGBoost is consistent with the wider application of data-driven methods to nonlinear construction-material relationships [23], [24], [31]. However, the present models are treated as exploratory proof-of-concept tools because the experimental dataset contains only eight independent batch-temperature conditions. Their role is to illustrate the integration of bacterial-mortar experiments with machine-learning prediction rather than to establish a universally applicable design model.

1.7 Research Gap

Previous studies have shown that bacteria-based self-healing concrete can improve crack healing, reduce permeability, and support strength recovery [3], [5], [6], [8], [9], [11], [18].

Many studies have also confirmed that calcium lactate is an effective nutrient source for bacterial self-healing concrete [14]–[17], [19]. However, there are still important research gaps.

First, limited studies have investigated the combined effect of calcium lactate concentration and temperature on *Bacillus subtilis* mortar. Many studies examine bacteria type, nutrient dosage, curing age, or crack width separately, but fewer studies combine nutrient optimization with temperature exposure. Second, most studies are conducted at room temperature or under one curing condition. This does not fully represent field conditions where concrete may face cold, normal, warm, or high-temperature exposure.

Third, there is limited understanding of the optimum nutrient-temperature interaction. Calcium lactate may support bacterial activity, but too much calcium lactate may not always improve strength or healing. Similarly, higher temperature may increase reaction rate, but very high temperature can reduce bacterial activity [16], [20], [21]. Therefore, the optimum condition must be experimentally studied.

Fourth, limited undergraduate-level experimental studies combine bacterial self-healing mortar with completed data-driven prediction. AI has been used for predicting concrete strength and bacterial concrete strength [23], [24], but its integration with temperature-sensitive bacterial self-healing data remains developing. This creates a clear need for research that connects experimental biomineralization with carefully bounded machine-learning prediction.

This research fills in these gaps by examining the effect of three different dosages of calcium lactate and four different temperatures on the *Bacillus subtilis* bacterial mortar. In addition, the work comprises compressive strength testing before and after healing, induced crack control testing, healing-efficiency testing, and preliminary predictions of healing efficiency using XGBoost within the experimental domain measured.

1.8 Problem Statement

Shrinkage, loading, temperature change, chemical attack, environmental exposure are common causes of cracks in concrete structures during their service life. These cracks permit harmful substances to penetrate into the concrete, lower the durability and increase

the risk of corrosion of reinforcement. Traditional repair techniques involve expensive, time-consuming and sometimes impractical processes that are difficult to implement in hard-to-reach locations. Meanwhile, with climate change, structures are more exposed to the aggressive environmental conditions, where durability and service-life extension becomes more important.

Bacteria-based self-healing concrete offers a sustainable solution by using biomineralization to produce calcium carbonate inside cracks. However, the performance of this system depends strongly on bacterial species, nutrient source, nutrient dosage, temperature, and curing condition. There is still limited understanding of the interaction between calcium lactate dosage and temperature in *Bacillus subtilis* mortar. Without this understanding, bacterial self-healing materials cannot be confidently adapted to different climate conditions. Therefore, this study investigates the effect of calcium lactate dosage and temperature on crack-healing efficiency and compressive-strength recovery of *Bacillus subtilis* mortar.

1.9 Research Aim

The aim of this research is to evaluate the self-healing performance of *Bacillus subtilis* bacterial mortar under different calcium lactate dosages and temperature conditions, identify the best-performing tested condition for crack-healing efficiency and compressive-strength recovery, and develop preliminary XGBoost models for early-condition prediction within the investigated range.

1.10 Research Objectives

The objectives of this study are:

To prepare *Bacillus subtilis* bacterial mortar cubes using a bacterial concentration of 10^8 cells/mL, a cement:sand ratio of 1:2, and a water–cement ratio of 0.45.

To investigate the effect of calcium lactate concentrations of 0.5%, 1.5%, and 10% on biomineralization-based crack healing.

To evaluate the effect of temperature conditions of $-15\text{ }^{\circ}\text{C}$, $20\text{ }^{\circ}\text{C}$, $40\text{ }^{\circ}\text{C}$, and $50\text{ }^{\circ}\text{C}$ on bacterial healing performance.

To determine compressive strength before crack induction and after the healing period.

To calculate healing efficiency and identify the best-performing tested calcium lactate-temperature condition.

To develop and evaluate preliminary XGBoost regression models for predicting Day-28 healing efficiency and post-healing compressive strength within the investigated experimental range.

1.11 Research Questions

This study addresses the following research questions:

How does *Bacillus subtilis* contribute to crack healing in cementitious mortar through biomineralization?

How do different calcium lactate concentrations affect crack-healing efficiency and compressive-strength recovery?

How do different temperature conditions affect the observed crack-healing and post-healing mechanical performance?

Which tested combination of calcium lactate dosage and temperature gives the best overall healing performance?

How can compressive-strength recovery and healing efficiency be used to evaluate bacterial self-healing mortar?

How effectively can a three-input XGBoost framework provide preliminary predictions of Day-28 healing efficiency and post-healing compressive strength within the investigated experimental range?

1.12 Scope of Study

This study is limited to laboratory-scale cement mortar cubes. The cube size used in the experiment is 50 mm × 50 mm × 50 mm. The mortar mix uses a cement:sand ratio of 1:2 and a water–cement ratio of 0.45. The bacterial species used is *Bacillus subtilis*, and the

bacterial concentration is 10^8 cells/mL. Calcium lactate is used as the nutrient source at 0.5%, 1.5%, and 10%. The temperature conditions are $-15\text{ }^{\circ}\text{C}$, $20\text{ }^{\circ}\text{C}$, $40\text{ }^{\circ}\text{C}$, and $50\text{ }^{\circ}\text{C}$.

The experimental work includes casting of mortar cubes, initial compressive strength testing, controlled crack induction, healing under selected temperature conditions, final compressive strength testing, and healing-efficiency assessment. The study focuses mainly on compressive-strength recovery and crack-healing performance. The research does not include reinforced concrete beams, long-term field exposure, chloride penetration testing, carbonation testing, freeze–thaw testing, or full-scale structural testing. SEM, XRD, permeability testing, and advanced image analysis are discussed in the literature, but they were not performed as part of the present experiment. The machine-learning component was completed as a preliminary XGBoost modelling exercise and remains limited to the measured experimental domain.

The completed machine-learning component is limited to the same experimental domain as the healing study: Batches B and C, calcium lactate concentrations of 0.5% and 1.5%, temperatures from $-15\text{ }^{\circ}\text{C}$ to $50\text{ }^{\circ}\text{C}$, and initial crack widths from 0.279 to 0.722 mm. It does not constitute external validation and should not be extrapolated to other mixtures, bacterial concentrations, specimen sizes, or environmental conditions without additional experimental data.

1.13 Significance of the Study

This study is significant for researchers because it adds experimental knowledge about the combined effect of calcium lactate concentration and temperature on *Bacillus subtilis* self-healing mortar. It helps explain why a moderate nutrient concentration may perform better than too little or too much nutrient. It also helps show how temperature can improve or reduce bacterial healing performance.

This study is highly useful for the construction industry in developing durable and low-maintenance cementitious materials. By optimizing bacterial self-healing systems, cracks caused deterioration, repair costs, and service interruptions can be reduced. The research offers valuable data for the durability engineer on the efficiency of compressive-strength

recovery and healing for a range of exposure conditions. This can help in developing the material for future applications of durability.

The research promotes the concept of extending service life and minimizing material usage for sustainable development of infrastructure. Environmental impact can be diminished by having long-lasting concrete structures that do not need to be repaired and/or replaced so often. The study supports the assessment of the behavior of the bacterial self-healing mortar in climate adaptation scenarios of cold temperatures, normal temperature, warm temperature and high temperature. Completed XGBoost models provide an indication of how early-condition variables can be used to screen materials for design and how much data is required before the model can be used for the design.

The finished predictive framework also illustrates the possibility of using experimental crack-healing data to create an early screening tool. The models are developed using only variables available at Day 0, and show the ability to compare the healing conditions of candidates prior to the final 28-day observation period and destructive compressive-strength test, but still clearly highlight uncertainty and data gaps.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Concrete is the most popular construction material due to its economic, availability of local resources, ease of casting and compression strength [32]. Yet concrete is also a brittle material and cracking is virtually inevitable throughout its service life [33]. Cracks can occur as a result of shrinkage, temperature fluctuations, mechanical loading, freeze–thaw, chemical attack, settlement, or inadequate construction techniques [34]. Small cracks do not necessarily lead to structural failure, but can provide direct access for water, oxygen, carbon dioxide, chloride ions and sulphate ions [35]. The harmful substances can decrease the durability and can lead to corrosion of steel reinforcement in reinforced concrete structures [18].

The cost, labour, time and construction material required for repair and maintenance of cracked concrete structures are very high [33]. In many cases, repair work is challenging as cracks can be present within concrete underground members, water-retaining structures, bridges, tunnels and marine structures [6]. Typical repair materials like epoxy, polymer grout, cement grout, and surface coatings can seal cracks that are visible, but need human inspection and manual application [36]. These repair techniques are also restricted as they may not be able to access cracks in the center of the members, or they may not be easily compatible with the cementitious matrix over the long term [34].

In response to the damaging impact of cracking and to enhance the durability of cementitious materials, self-healing concrete has been developed [32]. The concept of self-healing concrete is basically to close cracks automatically or semi-automatically without external repair works. Self-healing can be accomplished using autogenous methods, chemical admixtures, encapsulated agents, fibres, vascular systems or biological self-healing [32]. Of these, bacteria-based self-healing concrete is the method that has been the focal point of major research interest due to its ability to generate calcium carbonate within cracks by biomineralization [37].

This chapter reviews the literature related to bacteria-based self-healing concrete, biomineralization, Microbially Induced Calcium Carbonate Precipitation, *Bacillus subtilis*, calcium lactate, temperature effects, compressive-strength recovery, healing efficiency, sustainability, and Artificial Intelligence applications in cementitious materials. The review is directly linked with the experimental work of this thesis, which investigates *Bacillus subtilis* bacterial mortar using 0.5%, 1.5%, and 10% calcium lactate under -15°C , 20°C , 40°C , and 50°C temperature conditions.

2.2 Cracking and Durability Problems in Cementitious Materials

Cracking is one of the most common causes of durability loss in concrete and mortar [34]. Concrete has high compressive strength, but its tensile strength is relatively low [35]. When tensile stress exceeds the tensile capacity of the cementitious matrix, cracks begin to form [33]. These cracks may be microcracks or visible cracks depending on the loading, curing condition, material properties, and exposure environment [32].

Cracks reduce durability because they increase the permeability of concrete [38]. Higher permeability allows aggressive fluids and gases to enter the concrete more easily [18]. Once harmful agents enter the material, they may damage the cement matrix and accelerate reinforcement corrosion [35]. This can reduce stiffness, strength, serviceability, and long-term structural safety [34].

The presence of cracks is also important for climate adaptation because concrete structures are exposed to changing environmental conditions [39]. Temperature cycles, wetting–drying cycles, moisture variation, and aggressive exposure conditions may increase deterioration [40]. Therefore, improving the crack resistance and crack-healing capacity of cementitious materials is important for sustainable infrastructure [33].

Several studies show that the repair of cracked concrete can consume large economic resources [33]. Repair activities also increase environmental impact because they require additional materials, equipment, transport, and energy [41]. If concrete structures can heal small cracks at an early stage, the need for frequent repair can be reduced [42]. This is one of the main reasons why self-healing concrete is considered a promising material for durable and sustainable construction [32].

2.3 Concept of Self-Healing Concrete

Self-healing concrete refers to cementitious material that can close cracks and recover part of its original performance after damage [32]. The healing process may restore water tightness, reduce permeability, improve stiffness, or recover mechanical strength [34]. The healing response may be natural, stimulated, or engineered depending on the type of mechanism used [32].

Autogenous self-healing is the natural crack-healing ability of cementitious materials [35]. It occurs mainly due to continued hydration of unhydrated cement particles and precipitation of calcium carbonate inside cracks [35]. Autogenous healing is usually effective only for very small cracks and normally requires the presence of moisture [32]. The healing process also depends on crack width, age of concrete, water availability, and cement composition [35].

Autonomous self-healing uses additional healing agents to improve crack closure [32]. These healing agents may include bacteria, capsules, mineral admixtures, expansive agents, polymers, or vascular networks [36]. Autonomous healing is useful because it can be designed to activate when cracking occurs [33]. Bacteria-based self-healing is one type of autonomous healing in which microorganisms are used to precipitate calcium carbonate [37].

Biological self-healing is attractive because the healing product is usually calcium carbonate, which is compatible with cementitious materials [37]. Calcium carbonate can fill cracks, block pores, and reduce the transport of water and harmful ions [38]. The biological method is also considered environmentally friendly compared with many chemical repair systems [37]. For this reason, bacteria-based self-healing concrete has become a major research area in sustainable construction materials [43].

2.4 Biomineralization and MICP Mechanism

The phenomenon of mineralization by living organisms is known as biomineralization [37]. Calcium carbonate [38] is the most important biomineralization product in bacteria based self-healing concrete. Microbial Induced Calcium Carbonate Precipitation is a process of changing the chemical environment by the activity of microorganisms that results in the precipitation of calcium carbonate [37]. This process is also known as MICP [36].

The presence of calcium ions, carbonate ions, pH levels, dissolved inorganic carbon and nucleation sites are essential for MICP to occur [37]. The negatively charged functional groups on the bacterial cell surfaces can serve as nucleation sites [38]. These groups bind the calcium ions and promote crystallization [37]. When calcium carbonate starts to precipitate it can grow into cracks and pores [39].

Various microbial processes are available to precipitate calcium carbonate [37]. These pathways include ureolysis, denitrification, sulphate reduction, photosynthesis, and oxidation of organic calcium salts [38]. Ureolysis is one of the most investigated pathways as the urease-producing bacteria can easily produce the carbonate ions [37]. But ureolysis

may also result in the generation of ammonium as a by-product, which may not be desirable in all concrete applications [44].

Non-ureolytic pathways are also important for self-healing concrete [44]. In these pathways, bacteria metabolize calcium-containing organic compounds such as calcium lactate, calcium acetate, or calcium formate [39]. The metabolism of organic calcium salts can produce carbon dioxide, which can react with calcium hydroxide in the cement matrix to form calcium carbonate [30]. This pathway is important for the present thesis because calcium lactate was used as the nutrient and calcium source.

Calcium carbonate can occur in different crystal forms such as calcite, aragonite, and vaterite [39]. Calcite is usually considered the most stable calcium carbonate polymorph [39]. The morphology and amount of calcium carbonate can influence the quality of crack filling and the durability of repaired concrete [39]. Therefore, the success of bacterial self-healing depends not only on whether calcium carbonate forms, but also on how much forms and where it forms inside the crack [38].

2.5 Bacteria Used in Self-Healing Concrete

Many types of bacteria have been investigated for self-healing concrete [32]. The most common bacterial groups are from the genus *Bacillus* and the related genus *Sporosarcina* [37]. These bacteria are commonly selected because many of them can survive in alkaline environments and can form spores [44]. Spore formation is a key feature because concrete is a harsh environment with high pH, limited nutrients, low oxygen, and restricted pore space [44].

Spores are dormant bacterial structures that allow bacteria to survive unfavourable conditions [44]. When cracks form and water enters the concrete, spores can become active if nutrients and oxygen are available [45]. The activated bacteria then take part in biomineralization reactions and promote calcium carbonate precipitation [45]. This process allows cracks to be sealed from the inside [18].

Bacillus subtilis, *Bacillus sphaericus*, *Bacillus megaterium*, *Bacillus pseudofirmus*, *Bacillus cohnii*, *Bacillus halodurans*, and *Sporosarcina pasteurii* are examples of common bacteria species which are employed in self-healing concrete [44]. The mineralization

capacity, survival ability, urease activity, nutrient demand, and healing efficiency vary amongst species [46]. Thus, the choice of bacterial species is an important factor in the performance of bacterial concrete [46].

Numerous studies have confirmed that *Bacillus* species have a beneficial effect on the properties of concrete and mortar by precipitating calcium carbonate [38]. Bacterial treatment reported increases in compressive strength, decreases in porosity, permeability and crack filling for some studies [40]. The improvements are typically associated with the deposition of calcite in pores and cracks [38]. But, other factors also influence the performance of bacteria, such as concentration, carrier material, curing condition, nutrient type and environmental temperature [45].

2.6 *Bacillus subtilis* in Self-Healing Cementitious Materials

The species *Bacillus subtilis* is one of the most relevant bacterial species for this thesis as it was used in the experimental work. It is a spore forming bacterium which can endure harsh environment and can also be involved in the precipitation of calcium carbonate [9]. Its non-pathogenic nature and its suitability for using it in bio-based material applications make it a widely studied material. [30].

In the case of *Bacillus subtilis* mortar, the improvement of crack-healing ability and mechanical recovery have been reported by studies [9]. Yamasmit et al. examined the effect of *Bacillus subtilis* on different crack width in mortar with various curing conditions [9]. They concluded that bacterial mortar can enhance the crack-healing ability and mechanical properties of the concrete if suitable curing condition is applied to the cracks [9].

Nguyen et al. investigated the self-healing property of bacterial concrete with *Bacillus subtilis*, peptone and yeast extract [40]. They found that the addition of bacteria reduced the porosity and increased the strength, dynamic modulus, water uptake resistance, gas permeability and chloride resistance of their results [40]. Microstructural tests confirmed that the precipitated healing product inside cracks was calcium carbonate [40]. The study is of importance as it helps to correlate the self-healing ability of the bacteria with the mechanical properties and durability [40].

Ahmad et al. applied the spores of *Bacillus subtilis* in lightweight expanded clay aggregates, polyvinyl acetate fibres and air-entraining admixture [45]. They also found that the healing efficiency increased with increasing dose of bacteria and curing time [45]. They also employed ultrasonic pulse velocity, ImageJ surface crack analysis, SEM, XRD and FTIR to assess healing [45]. The present thesis also assesses the healing efficiency using crack observation and recovery of strength, making this paper extremely relevant.

Ahmad et al. compared the performances of *Bacillus subtilis*, *Bacillus sphaericus* and *Bacillus megaterium* in a bacterial concrete [46]. They found that bacteria behaved differently with respect to their ability to heal themselves, mechanical strength and freeze–thaw durability [46]. Additionally, the study revealed that *Bacillus subtilis* can aid in the healing process of calcite, although the overall healing effect might be affected by the carrier and exposure condition [46]. This highlights the importance of optimizing bacterial systems, and not relying on the assumption that all bacteria are the same.

Rahman et al. explored the self-healing property of *Bacillus subtilis* cement mortar by using local laboratory conditions [47]. They did a study on the compressive strength, crack recovery, water permeability, oxygen uptake, and bacterial activity [47]. When they tested the strength of cast bacterial mortar in appropriate chemical and bacterial solutions, they found that the specimens exhibited greater strength than the control specimens [47]. This work is directly related to the present thesis since it is about *Bacillus subtilis* cement mortar and improvement of strength.

2.7 Role of calcium lactate in Bacterial Self-Healing

To activate bacteria and promote the precipitation of calcium carbonate [16] nutrients are needed. The nutrients used in bacterial concrete are: calcium lactate, calcium acetate, calcium nitrate, calcium chloride and urea, yeast extract, peptone and other organic nutrients [39]. One of the most prevalent nutrient sources in non-ureolytic bacterial self-healing systems are calcium lactate [16]. It may be used as a source of calcium as well as an organic nutrient [19].

Calcium lactate is also crucial due to its potential to promote bacterial metabolic activity

and calcium carbonate formation [16]. Carbonate species can be generated and react with calcium ions to form calcium carbonate when bacteria are metabolizing lactate [30]. As it can fill cracks and decrease permeability in cementitious materials [19]. This mechanism is directly related to the present study since the calcium lactate concentrations of 0.5%, 1.5% and 10% were investigated.

Vijay and Murmu investigated the effect of calcium lactate on the compressive strength of microbial concrete and crack self-healing [16]. Their study demonstrated that dosage of calcium lactate has a significant impact on strength and healing response [16]. This is critical as the insufficient nutrient level can affect biomineralization, and if the nutrient is too high, it can interfere with the cementitious matrix [16]. They do find that it is necessary to find an optimum dose of calcium lactate as opposed to a fixed dose.

Muhammad et al. investigated the effect of the curing time and calcium lactate content on strength of *Bacillus subtilis* in a cement mortar [14]. They found that the compressive strength of the mortar containing bacteria and calcium lactate is better than that of the control mortar [14]. This paper is very relevant as it relates to cement mortar, *Bacillus subtilis*, calcium lactate, curing time and strength development [14].

Muhammad et al., also investigated the individual and combined effect of *Bacillus subtilis* and calcium lactate on cement mortar [15]. They found that the use of both bacteria and calcium lactate (CaL) was more effective than using either of them separately [15]. This reinforces the notion that bacterial cells and nutrient source should be suitably mixed to achieve good self-healing results [15].

Chaerun et al. applied bacteria to improve mortar properties and occurrence of self-healing by using calcium lactate pentahydrate [19]. The results indicated that calcium lactate pentahydrate could promote the formation of calcium carbonate and enhance the properties of mortar [19]. A peculiar aspect of this study is that it also employed the use of mortar and calcium lactate as part of a bacterial healing system [19]. It facilitates calcium lactate as a good candidate for the biomineralization process in cementitious materials.

Rahmawan et al. studied the use of *Bacillus subtilis* and calcium lactate for repairing concrete cracks [10]. They demonstrated high crack repair efficiency, improvement in the

compressive strength, decrease in the permeability and formation of calcite [10]. Presented one paper which strongly supports the presented thesis in the present because it brings together the combination of *Bacillus subtilis*, calcium lactate, crack repair, compressive strength, permeability, microscopy and image-based evaluation [10].

2.8 Nutrient Optimization and calcium lactate Concentration

The nutrient optimization and calcium lactate concentration are presented in the next section. One of the most important problems in bacterial self-healing mortar is optimization of nutrients [16]. If the nutrient content is not high enough, the bacteria could be suppressed, and calcium carbonate precipitation could be inadequate [14]. In the case of an excessive nutrient level, it can interfere with the hydration of the cement, the pore structure, setting behaviour or strength development [16]. Thus, the nutrient concentration is required to be carefully selected.

This study employed calcium lactate concentrations of 0.5%, 1.5% and 10%. A low calcium lactate dosage is found in the 0.5% concentration. The 1.5% is a moderate dosage. The 10% concentration is a high dosage. It has been found that medium calcium lactate concentrations are beneficial for both the bacterial activity and matrix stability [16].

Calcium lactate has a positive effect on the healing process due to its calcium content and the fact that calcium and carbonates are metabolized products [19]. But, in certain cases, more than calcium lactate does not necessarily mean better concrete properties [16]. Very high organic nutrient content has been found to affect the strength in some studies or to leave unreacted residues in the matrix [15]. This is why varying calcium lactate dosage happens to be necessary to determine the best dosing amount.

1.5% calcium lactate proved to be the best performance in this thesis. This is in line with the overall literature finding that a moderate nutrient level is better than a very high or very low nutrient level [16]. The discovery also reinforces the notion that the interaction between the bacteria, nutrient source and exposure condition should be taken into consideration for the optimization of bacterial concrete [39].

2.9 Bacterial Incorporation and Protection Methods

Bacteria can be added to concrete either directly or through carrier materials [48]. The mechanical mixing, high alkalinity, hydration heat and pore refinement result in a decrease in bacterial survival in direct addition, making it simple but not always the best choice. If there are no cracks, bacteria and nutrients can be encapsulated or immobilized [49]. Spores and nutrients also may be carried on carrier materials [45].

Expanded clay is one of the common carrier materials for bacteria in self-healing concrete [48]. Lucas et al. studied bacteria encapsulated in expanded clay and added to concrete by aggregate replacement [48]. Their work showed that the biological approach is a suitable method for self-healing, and expanded clay can serve as a bacterial carrier [48]. This is relevant because carriers can improve bacterial viability and long-term self-healing capacity.

Ahmad et al. reported that lightweight expanded clay aggregates can accommodate *Bacillus subtilis* spores and nutrient media for a long period [45]. They also found that PVA fibres can bridge cracks and provide nucleation sites for calcite precipitation [45]. This shows that bacterial healing can be improved by combining biological activity with physical crack-bridging mechanisms [45].

De Koster et al. investigated geopolymer coating of bacteria-containing granules for self-healing concrete [49]. Their study showed that coating can protect water-soluble healing agents during mixing and can later allow activation when cracks occur [49]. This method is important because premature nutrient leaching can reduce the effectiveness of bacterial self-healing systems [49].

The present thesis used bacterial mortar without focusing on advanced encapsulation systems. This creates a clear difference between the present study and several carrier-based studies. The direct use of *Bacillus subtilis* and calcium lactate can still provide useful information for basic optimization of nutrient concentration and temperature. However, future research may improve the system by incorporating suitable bacterial carriers.

2.10 Effect of Temperature on Biomineralization and Self-Healing

The temperature plays an important part in the environment of the bacterial self-healing mortar. It has a negative impact on the survival of bacteria, enzyme activity, metabolic rate, spore germination, and calcium carbonate precipitation [21]. The bacteria are biological, so low and high temperature can detrimentally affect performance if the bacteria become inactive or stressed [21]. So, temperature is one of the critical parameters of the climate adapted self-healing material.

Wang et al. investigated the temperature dependent MICP between 4 °C and 50 ° C [21]. They found that the bacterial density, bacterial activity, attachment and calcium carbonate precipitation are influenced by the temperature [21]. The study revealed that the reaction rates and the growth of bacteria can be decreased under low temperatures [21]. It has also demonstrated that high temperature can also inhibit bacterial activity when the temperature is above the tolerance of the bacteria [21].

Ahmad et al. explored the thermal sensitivities of microbial self-healing concrete at various temperatures with *Bacillus* strains [20]. They have compared the effectiveness of bacterial healing under room temperatures and below zero temperatures [20]. The results showed that temperature had an effect on crack healing, crack strength and mineral precipitation [20]. This underlines the significance of the study of -15 °C, 20 °C, 40 °C and 50 °C in the present thesis.

Bacterial metabolism is expected to be reduced at low temperatures at -15 °C, as the biological reaction rate will be reduced at low temperatures [21]. In addition, cold conditions may reduce water flow and calcium carbonate precipitation [20]. Thus, healing at -15 °C could be slow or incomplete based on moisture availability and the tolerance of the bacteria [20].

The bacterial systems are usually stable at 20° C as it is a convenient laboratory temperature [9]. The reason for conducting many self-healing concrete experiments at room temperature is that this temperature supports the curing process, and the activity of microorganisms and experimental control [9]. The optimum temperature for MICP, however, may not always be at room temperature [21].

If the bacteria are capable of tolerating the temperature, the metabolic activity and precipitation rate of the bacteria can be increased at 40 °C [21]. For the present thesis, it was determined that 1.5% calcium lactate at 40 °C gave the best performance observed. This may be the result of a balance between warm temperature, which facilitated biological or chemical healing processes, but not inducing excessive thermal stress. The interpretation is consistent with the study that MICP performance can be enhanced at moderate-high temperature and then be reduced at excessive temperature [21].

Bacterial activity might be affected by high temperatures at 50 °C, since high temperature can have an adverse effect on cells or enzyme activity [21]. If the temperature is too high it can also restrict moisture and result in poor curing conditions [20]. Thus, the decreased activity and biomineralization at 50 °C compared with 40 °C is an outcome of thermal stress.

2.11 Crack-Healing Efficiency Evaluation Methods

Typically, the efficiency of healing is determined by the decrease in crack width, crack area, permeability, water absorption, ultrasonic pulse velocity, or mechanical strength recovery [50]. Various techniques may yield different results, depending on what they are measuring [50]. Surface sealing is measured by visual crack closure, and functional sealing is measured by permeability testing [50]. Strength recovery is an increased mechanical improvement following damage [45].

The closure of cracks in self-healing concrete is usually studied using microscopy [45]. The width of surface cracks can be measured using optical microscopy before and after healing [50]. Images can be taken and crack area reduction can be quantified using image analysis software, e.g., ImageJ [45]. This approach is helpful since it offers a numerical healing efficiency based upon the visible closure of cracks [45].

The morphology and chemical composition of the healing products are observed using SEM and EDS [40]. SEM can provide information on whether a crystal has grown in the crack [40]. EDS can confirm the presence or not of Ca, C in the healing product [40].

These techniques have been used in order to ensure that cracking is due to a precipitation of calcium carbonate and not to loose particles alone [40]. XRD analysis can be performed

to determine the mineral phases in the sample including calcite, aragonite or vaterite [45]. Another use of FTIR is to assist in identifying carbonate-based healing products [45]. Such methods are crucial since self-healing of the bacteria requires mineralogical proof [45]. In some of the concrete bacterial studies, calcite has been found to be the primary self-healing product [46].

Permeability testing is significant because surface crack closure does not necessarily imply internal sealing of the crack [50]. Rooftop concrete samples tested in permeability tests by Van Mullem et al. demonstrated very good sealing properties when exposed to water [51]. This demonstrates that the functional healing can be assessed by measuring decreases in water flow across cracks [51].

The present thesis centers on the efficiency of healing and recovery of compressive strength. After incubation, image-based analysis is helpful to observe the crack closure. Prior to cracking and after healing, strength testing gives mechanical evidence of recovery. These techniques can then be used together to assess the **self-healing** performance both physically and mechanically.

2.12 Compressive Strength Recovery in Bacterial Concrete

Compressive strength is an important mechanical property of mortar and concrete. Bacterial self-healing can have two significant effects on the compressive strength. The first is that bacteria can precipitate and make the matrix denser [40]. Second, when cracks are induced, calcium carbonate may be formed inside the cracks, which will restore some of the lost strength [47].

Nguyen et al. found that the use of *Bacillus subtilis* in bacterial concrete demonstrated that the strength and durability parameters were enhanced [40]. The increase in the porosity was associated with better mechanical and durability characteristics [40]. This indicates that the internal structure of concrete can be enhanced by bacterial precipitation [40].

The results presented by Vijay and Murmu indicated that the calcium lactate has an impact on compressive strength and crack healing in microbial concrete [16]. The results of their research are relevant to the current thesis, as the concentration of calcium lactate in the past thesis is also analyzed, along with the ability to recover its strength. The nutrient to strength

relationship is not necessarily linear, since too much nutrient could be a drawback when its effect is on the matrix [16].

Muhammad et al. demonstrated that the use of *Bacillus subtilis* and calcium lactate can help increase the strength of the mortar if the curing time and the amount of calcium lactate are optimised [14]. This reinforces the present thesis, since mortar cubes were used for the experimental work, while large concrete members were used for the theoretical work. In particular, studies carried out in mortars can be very helpful to understand the fundamental performance of bacterial cementitious materials [14].

Ahmad et al. studied the compression strength recovery of preloaded bacterial concrete cube samples [46]. Their study demonstrated that the mixes containing bacteria had better strength restoration and freeze–thaw durability than the control mixes [46]. This study shows that self-healing ability of bacteria after damage is an important factor for strength recovery [46].

In this thesis, compressive strength was determined prior to inducing cracks and following healing. The procedure enables the strength recovery of bacterial healing to be calculated. The observed optimum at 40 °C and 1.5% calcium lactate indicates that the quantity of healing products formed as well as their efficacy, was nutrient concentration and temperature dependent.

2.13 Durability Improvement through Bacterial Self-Healing

The bacteria are capable of repairing cracks in cement. The bacteria can heal cracks in cement. The primary function of self-healing concrete is to improve its durability. If the repaired crack is permeable or weak, it cannot be considered to have been closed unless additional measures are taken [50]. Proper bacteria healing should minimize water absorption, gas permeability, chloride migration, and subsequent damage [40]. Hence, many bacterial concrete studies have included durability testing as an important component of their investigations [40].

Nguyen et al. found that bacterial self-healing concrete decreased water uptake, gas permeability and chloride permeability [40]. The result is significant because the mobility of fluids and ions in the concrete matrix has a high influence on the durability [40].

Bacterial calcium carbonate can lead to the blocking of pathways for aggressive agents when it fills cracks [38].

Rahmawan et al. demonstrated that *Bacillus subtilis* and calcium lactate were able to reduce permeability, and enhance the effectiveness of crack repair [10]. Their work also confirmed that calcite formation took place, using microscopy [10]. It is consistent with the concept that bacterial crack healing has a positive effect on both surface closure and functional durability [10].

Ahmad et al. showed that bacterial concrete can improve freeze–thaw durability after healing [46]. Freeze–thaw action is a major durability concern in cold climates because water inside pores and cracks can expand during freezing [46]. Bacterial crack filling can reduce water movement and help improve resistance to this type of damage [46].

The present thesis includes temperature conditions down to $-15\text{ }^{\circ}\text{C}$. This is relevant to durability because cold environments can reduce bacterial activity and affect crack healing. Understanding bacterial performance under different temperatures can support the development of self-healing cementitious materials for different climate conditions.

2.14 Sustainability and Life-Cycle Benefits

Self-healing concrete supports sustainability mainly through service-life extension and reduced repair demand [42]. Concrete production consumes large amounts of raw materials and contributes to carbon emissions through cement manufacturing [41]. Therefore, increasing concrete durability can reduce the need for replacement and repair materials [42].

Al-Gheethi et al. analysed the life-cycle assessment of bio-cementitious material production [41]. Their study showed that cement contributes strongly to global warming, human health damage, ecosystem damage, and other environmental impacts [41]. This means that any technology that reduces cement consumption or extends concrete service life can support sustainability [41].

Panza Uguzzoni et al. studied concrete self-healing from a life-cycle cost perspective [42]. Their work showed that self-healing materials can reduce maintenance costs, increase service life, improve residual value, and reduce environmental impacts [42]. This supports

the argument that self-healing concrete should be evaluated not only by initial cost, but also by long-term performance [42].

Li and Herbert mentioned the concept of self-healing concrete as a solution for sustainable infrastructure [33]. They pointed out that the self-healing system should be strong, secure, repeatable, and applicable to actual structures [33]. This is significant as the material that works best in a lab may not be sustainable in the field [33].

The first large-scale application of self-healing concrete using bacteria was investigated and published in Belgium, by Van Mullem et al. [51]. Their study indicated that it is feasible to put this on a large scale, but it is hard to verify this in the field without cracking while under observation [51]. This indicates that further research is needed to establish laboratory performance with realistic field conditions [51].

Mors and Jonkers analyzed the full-scale demonstrator projects of bacterial self-healing concrete [52]. In demonstrator projects, they found that this was safe to add bacterial healing agents into concrete, but proving the healing agents performed better under field conditions was still difficult [52]. This confirms that practical implementation requires standard testing, field monitoring, and long-term validation [52].

2.15 Artificial Intelligence and Machine Learning in Self-Healing Concrete

Using Artificial Intelligence, the strength, healing efficiency, crack closure, and durability performance of self-healing concrete can be predicted [23]. The bacterial concrete is a complex product and there are a lot of variables such as bacterial type, bacterial concentration, nutrient concentration, water–cement ratio, curing time, crack width, temperature, carrier type and condition of exposure [24]. These factors are inter-related and non-linear [23]. Machine learning algorithms may be used to recognize trends in experimental data, and to decrease testing of trial mixes [23].

Almohammed et al. assessed soft computing techniques for predicting the compressive strength of bacterial concrete [23]. Their study compared different models and showed that machine learning can predict bacterial concrete strength using material and curing variables

[23]. This is directly relevant because the present thesis measured compressive strength before and after healing.

Vijay and Murmu applied Artificial Neural Networks to predict microbial concrete compressive strength [24]. Their model used calcium lactate dosage, bacterial concentration, and age as input variables [24]. This is highly relevant because the present thesis also uses *Bacillus subtilis*, calcium lactate concentration, and strength testing [24].

Theja et al. used an Artificial Neural Network to predict the strength properties of self-healing concrete [53]. Their study showed that ANN predictions can match experimental strength data when suitable input variables are used [53]. This supports the cautious use of AI models for bounded prediction of bacterial mortar performance [53].

Cao et al. developed optimized machine learning models to predict self-healing efficiency in recycled aggregate concrete [31]. Their study identified crack width as a highly influential factor in healing performance [31]. This is useful because crack size is one of the main parameters controlling healing efficiency in cementitious materials [31].

Onyelowe et al. used metaheuristic machine learning methods to model the influence of bacteria concentration on mechanical properties of self-healing concrete [54]. Their work showed that optimized algorithms can predict slump, flexural strength, and compressive strength with high accuracy [54]. This supports the use of advanced AI models for bacterial concrete optimization [54].

De Oliveira and Stefani studied synthetic data for machine learning prediction of the bacterial concrete self-healing capacity [55]. They found in their study that synthetic data can enhance model training, when the experimental data is limited [55]. This is crucial as self-healing concrete datasets are typically costly and limited in volume [55].

Vedrtnam et al. used finite element simulation to model bacterial self-healing in concrete using microstructural transport and precipitation modelling [56]. Their model simulated diffusion, reaction kinetics, calcium carbonate precipitation, and crack closure efficiency [56]. This indicates that expanded data-driven frameworks may eventually combine experimental data, image analysis and numerical simulation, provided they are supported by larger verified datasets [56].

Meghashree et al. combined *Bacillus subtilis* self-healing concrete with Random Forest and Kuhn-Tucker optimization [30]. Their study reported complete crack healing under optimal curing conditions and used modelling methods to maximize healing efficiency [30]. This paper is especially relevant to the present thesis because it combines *Bacillus subtilis*, calcium lactate, healing efficiency, and AI-based optimization [30].

Milović et al. used Artificial Neural Network modelling to study compressive strength improvement in cement containing individual and co-culture *Bacillus* bacteria [57]. Their study showed that ANN can support the analysis of bacterial cement strength under different environmental conditions [57]. This supports the AI-driven part of the present thesis because bacterial activity and strength recovery depend on several interacting variables [57].

2.16 Integration of AI with Bacterial Self-Healing Concrete

The incorporation of AI at various stages with bacterial self-healing concrete is possible. It can be utilized in mix design, experiment planning, crack monitoring, healing efficiency calculation, strength prediction and durability assessment. The variables of the experiments in the present thesis are: Calcium lactate concentration, temperature, bacterial concentration, initial compressive strength, final compressive strength, and healing efficiency. These variables can be used as inputs and outputs of future machine learning models.

In this study, a simple AI framework can be used with the inputs of calcium lactate concentration and temperature. Alternatively, initial compressive strength can be inputted. Outcome variables may be healing efficiency and recovery of compressive strength. Random Forest can be helpful if it can detect nonlinear relationships and be used in identifying which input variables are important. ANN can be beneficial if there is more experimental data to learn from when learning complex patterns. If a large image database is created, deep learning can be applied in the future for the analysis of the crack image.

There is also the option of image-based AI for automation of crack measurement. Images of the crack prior and after healing can be analyzed with the image processing software (ImageJ) or computer vision techniques. Numerical data can be obtained for the crack area,

crack width and crack closure. The values can then be coupled with a temperature measurement and a nutrient measurement to make a prediction of the efficiency of healing. This method can minimize the chance of human error and provide greater repeatability.

AI integration is beneficial for climate adaptation due to the variations in temperature and environmental factors that concrete structures undergo. A trained model can be used to predict the bacteria-nutrient combination that will work best in a particular climate. For instance, in the present thesis, it was found that 1.5% calcium lactate and 40 °C worked best. If there are more data points, AI could forecast if it is appropriate for other curing ages, crack widths, humidity and bacterial concentrations.

2.16.1 Relevance of XGBoost to the Present Study

The literature that was surveyed in Section 2.15 and 2.16 showed that machine learning can be used to support the prediction of the strength and behaviour of the bacteria concrete [23, 24, 31, 53, 54]. The aim of the present study is to extend this concept by building two regression models by means of XGBoost, a gradient-boosted decision-tree method, using the measured data from the *Bacillus subtilis* mortar.

Healing temperature, calcium lactate concentration and initial crack width were chosen as the selected inputs. The exposure condition that affects the chemical and bacterial activity is the temperature factor, the nutrient dosage is the concentration of calcium lactate, and the initial damage state is the Day-0 crack width. The results obtained were healing efficiency after 28 days, and compressive strength after healing. This arrangement offers a forecast of the condition of the crack before it is subjected to the direct target leakage that would be required if the Day-28 crack width was used to predict the Day-28 healing efficiency.

Past studies also indicate that self-healing datasets have often been small, as each observation has involved specimen preparation, controlled damage, prolonged exposure to healing temperatures, analysis, and mechanical testing [55]. Therefore, a model-validation strategy is of critical importance. For the 8 independent healing-efficiency records, leave-one-out validation is employed, and for the post-healing strength data, leave-one-

condition-out validation is used, to prevent dividing replicate specimens across the different batch-temperature conditions.

2.17 Critical Review of Previous Studies

Self-healing concrete with bacteria has been reported in literature as a promising technology for enhancing the durability and minimizing damage caused by cracking [32]. Numerous studies have shown that the main healing process in bacterial concrete is calcium carbonate precipitation [37]. Research also indicates that *Bacillus* species can be utilized because they can form spores and can survive extreme environments of concrete [44]. The use of *Bacillus subtilis* is well supported since it has been proven that it can enhance the strength recovery, crack healing and durability of mortar and concrete [9].

It has also been shown in the literature that calcium lactate is a good nutrient source for bacterial self-healing systems [16]. It has been found that calcium lactate can enhance crack closure and strength when applied in conjunction with appropriate bacteria in several studies [14]. But due to this, the concentration of nutrients should be optimized as lack of nutrients can inhibit healing while excess nutrient can decrease performance [16]. This directly supports the present study of 0.5%, 1.5% and 10% of calcium lactate.

Another variable is temperature, which although important, is less well known. MICP research indicates that the bacteria activity and/or calcium carbonate precipitation can be influenced by temperature [21]. Recently some studies have focused on bacterial concrete at cold and room temperature [20]. But, there were fewer studies involving multiple concentrations of calcium lactate and multiple temperatures for *Bacillus subtilis* mortar. This is one of the main gaps addressed by the present thesis.

In previous studies, various healing evaluation methods also are used. Some studies employ the visual closure of the cracks, others permeability, SEM, XRD, UPV, or strength recovery [50]. This is not a straightforward comparison as it is determined by the test method [50]. In the present thesis healing efficiency and compressive-strength recovery, which gives visual, as well as mechanical evaluation, are used.

Though limited, the literature on AI and bacterial self-healing concrete is growing. Some of the studies forecast compressive strength and others forecast healing efficiency or

optimise BC variables [23]. But, there is still a need for more data for experimentation to make reliable predictions [55]. The data presented in this thesis is then thus only a small proof-of-concept, which should be extended prior to practical use of the model.

2.18 Summary of Literature Review

According to the literature reviewed, one of the main durability issues with concrete is cracking as it can lead to the entry of harmful substances into the cementitious matrix [35]. Self-healing concrete has been introduced to prevent deterioration caused by cracks and to increase the service life [32]. Among the most promising methods is bacteria-based self-healing, which involves the in situ precipitation of calcium carbonate within cracks, using biomineralization [37].

Bacillus subtilis is an appropriate bacterium for self-healing mortar because it has the capacities of forming spores and withstanding harsh conditions, and of promoting the precipitation of calcium carbonate [9]. Calcium lactate is an important nutrient since it may supplement calcium and help maintain bacterial metabolism [16]. However, optimum calcium lactate concentration is required as high and low levels have an adverse effect on performance [16].

Bacterial activity and the efficiency of the MICP are strongly affected by temperatures [21]. Low temperatures may inhibit the bacteria's metabolism and very high temperatures may decrease the viability of bacteria [20]. There is literature to support that moderate-high temperature can aid in healing within the tolerable limits for the bacteria [21]. This is consistent with the present experimental result, which showed that the highest performance of the system was obtained with calcium lactate 1.5%, at a temperature of 40 °C.

It can be seen that healing efficiency can also be assessed either by image analysis, microscopy, permeability, strength recovery or by all the above methods [50]. The final preliminary models included in this thesis show a relationship between machine learning and compressive-strength recovery and healing-efficiency data [23]. The thesis is thus placed at the crossroads between the areas of biomineralization, enhancement of durability, sustainability, climate adaptation and data-based evaluation of materials.

CHAPTER 3 METHODOLOGY

3.1 Introduction

The experimental program was designed to understand the effect of *Bacillus subtilis*. The influence of and of calcium lactate dosage on the crack-healing and compressive-strength behavior of 1:2 cement–sand mortar. Bacteria-based healing in cementitious materials is commonly associated with microbially induced carbonate precipitation, through which mineral deposition may contribute to crack closure [7], [58]. The program included preparation of a bacterial suspension, production and water curing of mortar cubes, pre-crack compressive-strength testing, controlled crack induction, temperature-controlled healing, image-based crack monitoring, and post-healing compressive-strength testing.

The methodology was organized into the materials and experimental variables, bacterial-culture preparation, mortar mixing and casting, curing, initial strength testing, crack induction, healing exposure, image analysis, and calculation of healing and strength-recovery indices. All numerical values and procedures presented in this chapter were retained from the experimental record supplied for the study.

3.2 Research Design and Experimental Program

3.2.1 Experimental Groups

Four mortar groups were prepared. The control group was plain mortar and contained neither bacteria nor calcium lactate. Batches A, B, and C contained a *Bacillus subtilis* suspension with a target concentration of 10^8 cells/mL. Calcium lactate was incorporated as the bacterial nutrient at 10%, 0.5%, and 1.5% by mass of cement in Batches A, B, and C, respectively. Calcium lactate dosage was the primary treatment variable, while the recorded difference in its physical form was also retained in the methodology.

Table-3- 1 Control and experimental mortar groups

Group	Bacteria	Calcium lactate (% by mass of cement)
Control	None	0
Batch A	<i>B. subtilis</i> , 10 ⁸ cells/mL	10
Batch B	<i>B. subtilis</i> , 10 ⁸ cells/mL	0.5
Batch C	<i>B. subtilis</i> , 10 ⁸ cells/mL	1.5

3.2.2 Experimental Variables

The independent variables were the calcium lactate dosage and the healing temperature. The investigated healing temperatures were -15°C , 20°C , 40°C , and 50°C . The control variables included the cement–sand ratio, water–cement ratio, cube dimensions, bacterial concentration, proportion of bacterial suspension, curing duration, crack-induction procedure, image-acquisition schedule, and compressive-loading rate. The measured responses were crack width, percentage crack-healing efficiency, compressive strength, and percentage strength recovery.

3.2.3 Specimen Allocation

For the pre-crack compressive-strength assessment, three specimens were tested from each of the control group and Batches A, B, and C, giving 12 pre-crack specimens. The temperature-controlled healing phase included Batches B and C only. Three replicate specimens from each selected batch were assigned to each temperature condition and were subsequently tested after the 28-day healing period. The same cracked specimens that were monitored using ImageJ were used for post-healing compressive-strength testing. Batch A was excluded from the temperature-controlled healing phase; the experimental observations associated with that decision were reserved for Chapter 4.

Table-3- 2 Specimen allocation and testing stages

Experimental stage	Groups included	Replicates
Pre-crack compressive-strength test	Control, A, B, and C	3 specimens per group
Temperature-controlled crack-healing study	B and C	3 specimens per batch at each temperature
Post-healing compressive-strength test	B and C	The same 3 healed specimens per batch at each temperature

3.3 Materials

3.3.1 Cement

Ordinary Portland cement was used as the binder in all control and bacterial mortar mixtures. The same cement source was used throughout the experimental program to maintain consistency among the batches.

3.3.2 Fine Aggregate

Natural sand was used as the fine aggregate. The sand was sieved, and the fraction passing the No. 16 sieve (1.18 mm) and retained on the No. 30 sieve (600 μm) was selected. The same graded fraction was used for all specimens.

3.3.3 Mixing and Curing Water

Water was used for mortar mixing and for immersion curing. For bacterial mixtures, the bacterial suspension replaced 10% of the calculated mixing water by mass. The suspension therefore formed part of the total water corresponding to the specified water–cement ratio rather than being added as extra liquid.

3.3.4 Calcium Lactate

Calcium lactate was used as the nutrient source in the bacterial mortar. Its dosage was calculated as a percentage of cement mass.

3.3.5 Bacterial Culture and Nutrient Medium

An agar-plate culture of *Bacillus subtilis* was supplied by the Pakistan Council of Scientific and Industrial Research (PCSIR), Peshawar. Liquid nutrient broth was used as the growth medium for preparation of the bacterial suspension.

3.4 Preparation of the bacterial suspension

3.4.1 Sterilization of the Medium and Glassware

The nutrient broth and laboratory glassware were sterilized by autoclaving at 121 °C and 15 psi for 15 min. This exposure condition is specified for preparation of nutrient broth in the manufacturer's technical guidance [59]. After sterilization, the broth was allowed to cool to room temperature before inoculation.

3.4.2 Aseptic Transfer of the Parent Culture

The inoculating loop was heated in a flame until red hot and was then allowed to cool briefly. A colony of *Bacillus subtilis* was aseptically collected from the agar plate supplied by PCSIR Peshawar and transferred into the sterile nutrient broth. The inoculation was performed carefully to minimize contamination of the culture.

3.4.3 Incubation and Concentration Adjustment

The inoculated nutrient broth was placed in a shaking incubator for 24 h. A turbid bacterial culture was obtained after incubation. Optical-density measurement was used to prepare a suspension corresponding to a bacterial concentration of 10^8 cells/mL. This suspension was used in Batches A, B, and C as a 10% replacement of the total mixing water by mass.



Figure 3- 2 Parent culture of *Bacillus subtilis* on a nutrient agar plate.



Figure 3- 1 Sterilized liquid nutrient broth prepared for bacterial inoculation.



Figure 3- 5 Aseptic transfer of a *Bacillus subtilis* colony into nutrient broth.



Figure 3- 4 Shaking incubator used for preparation of the bacterial suspension.



Figure 3- 3 Autoclave used for sterilization of nutrient medium and laboratory glassware.

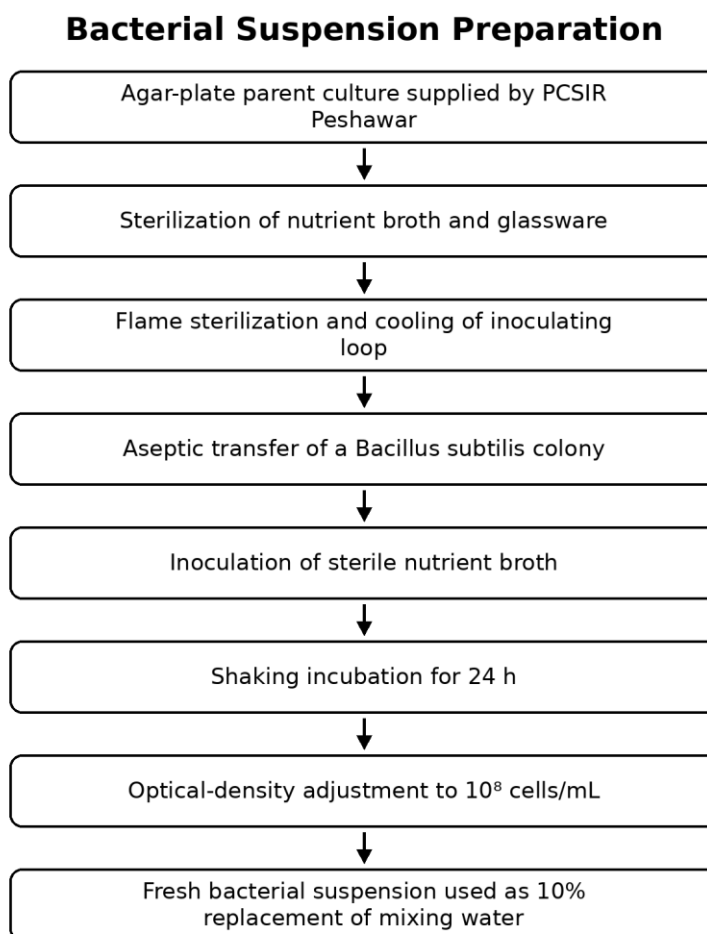


Figure 3- 6 Flowchart for preparation of the *Bacillus subtilis* suspension.

3.5 Mortar Mix Design and Specimen Preparation

3.5.1 Base Mix Proportions

All groups were produced using the same base mortar proportions. Cement and sand were proportioned at 1:2 by mass, and the water–cement ratio was maintained at 0.45. The specimens were cast as 50 mm × 50 mm × 50 mm cubes. The bacterial suspension used in the experimental batches had a concentration of 10^8 cells/mL and replaced 10% of the mixing water by mass.

Table-3- 3 Base mortar proportions and specimen characteristics

Parameter	Specified value
Cement:Sand ratio	1:2 by mass
Water-cement ratio	0.45
Cube dimensions	50 mm × 50 mm × 50 mm
Bacterial concentration in suspension	10 ⁸ cells/mL
Bacterial suspension	10% replacement of mixing water by mass

3.5.2 Manual Mixing and Casting

The cement and graded sand were first dry mixed manually until a uniform blend was obtained. Calcium lactate was then added at the dosage assigned to the relevant experimental batch. For Batches A, B, and C, the prepared bacterial suspension was combined with the remaining mixing water so that the suspension replaced 10% of the total mixing water by mass. The liquid was added gradually while manual mixing continued until a homogeneous mortar was obtained.

The fresh mortar was placed into 50 mm cube molds in two layers. Each layer was compacted to reduce entrapped air. After the final layer had been placed, the specimen surfaces were levelled and finished. The control specimens were prepared through the same procedure but without bacterial suspension or calcium lactate.

3.5.3 Demolding and Water Curing

After initial setting and hardening the cast specimens were left in the mold for 24 hours at room temperature. These molds were then carefully removed, to prevent damage to the edges and corners of the specimens. Immediately after demolding the cubes were wholly submerged in clean water, which was kept at about 23–27 °C. Pre-crack testing and crack induction were conducted after 28 days of keeping the specimens fully submerged. The curing sequence for the study is shown in figure 3-7.

Mortar Specimen Curing Sequence

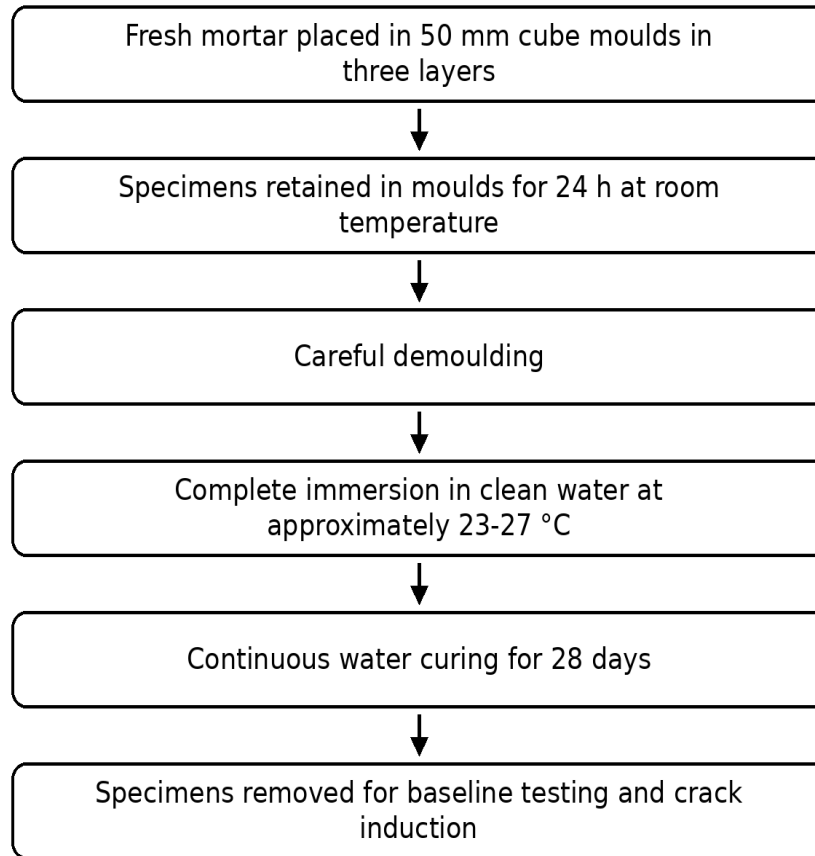


Figure 3- 7 Casting, demolding, and water-curing sequence for mortar cubes.



Figure 3- 8 Sieving of sand.



Figure 3- 9 Addition of calcium lactate.

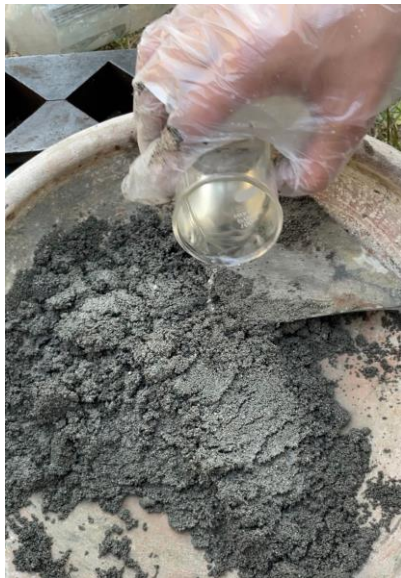


Figure 3- 10 : Bacterial suspension
Addition.



Figure 3- 11: Bacterial specimens

3.6 Pre-crack Compressive-Strength Test

3.6.1 Test Specimens and Preparation

Pre-crack compressive strength was determined after the 28-day water-curing period and before crack induction. Three 50 mm cubes from each of the control group and Batches A, B, and C were tested, producing 12 pre-crack measurements. Before testing, each specimen was visually inspected, its dimensions were verified, and loose particles were removed from the bearing surfaces.

3.6.2 Loading Procedure

Each cube was positioned centrally between the loading platens of a calibrated Universal Testing Machine (UTM). Compressive load was applied continuously and without shock at 1.5 kN/s until specimen failure. The maximum load sustained at failure was recorded. The use of 50 mm mortar cubes and the general compressive-strength procedure were based on ASTM C109/C109M-23 [60].

3.6.3 Compressive-Strength Calculation

The compressive strength of each specimen was calculated from the maximum failure load and the loaded cross-sectional area using Equation (3-1).

$$f = P / A$$

Equation 3-1

where f was the compressive strength in MPa, P was the maximum applied load at failure in N, and A was the loaded cross-sectional area in mm².

For the 50 mm cube specimens, the loaded area was calculated as shown in Equation (3-2).

$$A = 50 \text{ mm} \times 50 \text{ mm} = 2500 \text{ mm}^2$$

Equation 3-2

The individual results and the arithmetic mean for each group were retained as the pre-crack values for subsequent strength-recovery analysis.

3.7 Controlled Crack Induction

3.7.1 Crack-Induction Arrangement

Controlled cracking was performed after completion of the 28-day curing and pre-crack-strength program. A 1 mm diameter steel wire was positioned centrally beneath each cube before loading in the UTM. The wire created a narrow line of contact and acted as a stress concentrator so that cracking initiated at a controlled location. Controlled crack formation is important in self-healing studies because initial crack geometry can influence the observed closure response [61].

3.7.2 Loading and Stopping Criterion

Each specimen was centered on the loading platform with the wire aligned beneath the intended crack location. Load was applied at 1.5 KN/s. Loading was stopped immediately when the first visible surface crack appeared. The specimens were not loaded to complete failure during this stage. This approach preserved the cubes for subsequent healing exposure, image monitoring, and post-healing strength testing.

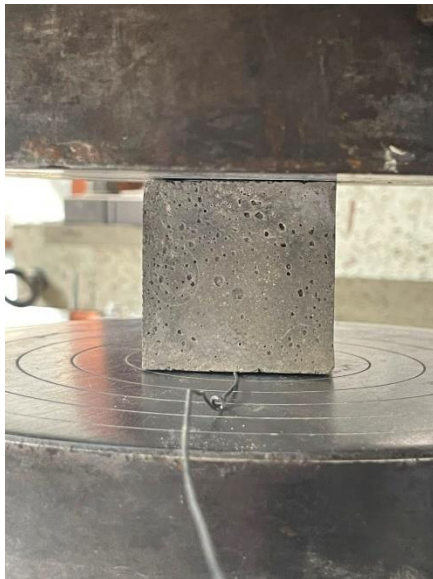


Figure 3- 12 : 1 mm wire placed under the specimen.

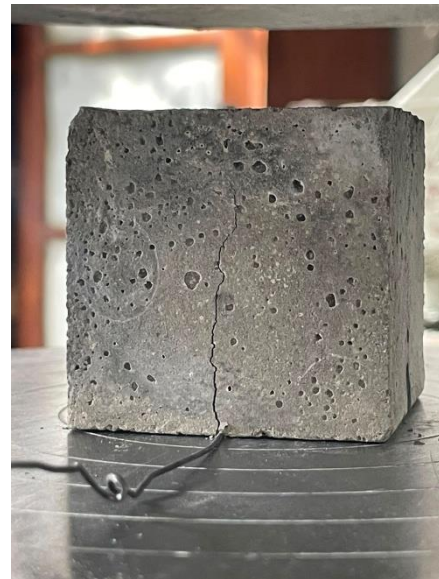


Figure 3- 13 :Controlled crack induction.

3.7.3 Day 0 Documentation

Immediately after crack induction, each specimen was labelled according to batch and intended temperature condition. A Day 0 photograph was captured using an iPhone 12. The camera position, crack location, lighting, and magnification were maintained as consistently as practicable for all subsequent images. The initial crack width was measured at a fixed-point using ImageJ software, and the same point was used for later measurements.

3.8 Temperature-Controlled Healing

3.8.1 Selection of Batches

Only Batch B, containing 0.5% calcium lactate, and Batch C, containing 1.5% calcium lactate, were included in the temperature-controlled healing *program*. Batch A was excluded from this phase. Experimental observations associated with that selection were reported in Chapter 4 rather than in the methodology.

3.8.2 Healing Environments

Cracked specimens were exposed to four temperature conditions for 28 days: -15°C , 20°C , 40°C , and 50°C . The -15°C condition was provided using a freezer. The 20°C , 40°C , and 50°C conditions were maintained using incubators provided by the Institute of Biotechnology and Genetic Engineering, The University of Agriculture Peshawar. The healing environments were maintained at approximately 95% relative humidity.

Table-3- 4 Temperature conditions used during the 28-day healing period

Condition	Temperature	Equipment	Relative humidity
Sub-zero	-15°C	Freezer	Approximately 95%
Room temperature	20°C	Incubator	Approximately 95%
Elevated temperature	40°C	Incubator	Approximately 95%
High temperature	50°C	Incubator	Approximately 95%

3.9 Crack Monitoring and Healing Evaluation

3.9.1 Image-Acquisition Schedule

The induced cracks were monitored throughout the 28-day healing period. Digital photographs were recorded on Day 0, Day 7, Day 14, Day 21, and Day 28. For each specimen, the same crack location was photographed using the same iPhone 12 and a consistent camera position and magnification. The observation schedule is summarized in Table 3-5.

Table-3- 5 Crack-monitoring schedule

Observation	Timing	Purpose
Day 0	Immediately after crack induction	Initial crack-width measurement
Day 7	7 days of healing	Interim crack-width measurement
Day 14	14 days of healing	Interim crack-width measurement
Day 21	21 days of healing	Interim crack-width measurement
Day 28	28 days of healing	Final crack-width measurement

3.9.2 ImageJ Measurement Procedure

Crack-width measurements were performed using ImageJ, an open-source scientific image-analysis program [62]. For each image, the measurement scale was established using a known reference dimension visible in the image. Crack width was then measured in millimeters at the same fixed point used for the Day 0 measurement. Maintaining one measurement location reduced variation caused by changes in crack geometry along its length.

3.9.3 Crack-Healing Efficiency

The percentage crack-healing efficiency at each observation time was calculated from the reduction in crack width using Equation (3-3).

$$\text{Healing Efficiency (\%)} = \frac{W_0 - W_t}{W_0} \times 100$$

Equation 3-3

where w_0 was the initial crack width measured on Day 0 and w_t was the crack width measured at the selected healing time. This method was represented as a fraction of the amount of cracking that occurred before and after the healing process, and was similar to the image-based approach in the evaluation of healing in bacterial systems [61].

3.10 Post-Healing Compressive-Strength Test

3.10.1 Specimen Preparation and Loading

The specimens in Batches B and C that were monitored were taken out of their environments at the end of the 28-day healing period. Excess surface moisture was removed, loose particles were cleaned up and the specimens were visually examined. The loading of each cube was carried out between the platens of the calibrated UTM, with the load applied at a rate of 1.5 kN/s until total failure. The maximum load was obtained, and the calculated compressive strength was done by applying the area of loading applied on the specimen (which was 2500 mm²) and the equation (3-1). The same equipment, specimen orientation and loading procedure were employed for the pre-crack compressive strength tests to enhance the comparability [60].

3.10.2 Strength-Recovery Calculation

Post-healing mechanical response was assessed by the mean compressive strength of the healed specimens and the mean pre-crack compressive strength of the companion batch of specimens. Percentage strength recovery was calculated using Equation (3-4).

$$\text{Strength Recovery (\%)} = \frac{f_h}{f_i} \times 100$$

Equation 3-4

where f_h was the mean compressive strength after the 28-day healing period and f_i was the mean pre-crack compressive strength obtained from the companion specimens of the same batch before crack induction.

3.11 Data Collection and Analysis

3.11.1 Recorded Data

The recorded data comprised the specimen identification code, batch designation, calcium lactate dosage, healing temperature, observation day, crack width, maximum compressive load, and calculated compressive strength. Day 0 images established the initial crack condition, while the Day 7, 14, 21, and 28 images documented crack-width development during healing.

3.11.2 Descriptive Analysis

Individual compressive-strength measurements were retained for each tested specimen, and the arithmetic mean was calculated for each experimental group. Crack-healing efficiency was calculated for each observation interval using Equation (3-3). Strength recovery was calculated from the relevant group means using Equation (3-4). The resulting crack-closure and mechanical-performance data were compared across calcium lactate dosages and healing temperatures in Chapter 4.

3.11.3 Machine-Learning Dataset Construction

The machine-learning dataset was assembled only from the measured records reported in Chapter 4. No synthetic, interpolated, or regression-generated observations were used for model training. The Day-28 healing-efficiency dataset contained eight independent records corresponding to Batch B and Batch C at -15 °C, 20 °C, 40 °C, and 50 °C. Each record represented one complete crack-monitoring sequence.

The post-healing compressive-strength dataset contained 24 specimen-level measurements, comprising three replicate strength results for each of the eight batch-temperature conditions. Although 24 strength values were available, the input variables were identical for the three specimens within each condition. The data therefore represented eight unique predictor combinations, and this grouping was retained during validation.

3.11.4 Model Inputs and Prediction Targets

Two independent regression models were developed. Both models used healing temperature, calcium lactate concentration, and Day-0 crack width as input variables. The

first target was Day-28 healing efficiency, while the second target was post-healing compressive strength in MPa. The input and output structure is summarized in Table 3-6.

Table-3- 6 Input and target variables used in the completed machine-learning models

Model	Input variables	Prediction target	Experimental records
Day-28 healing-efficiency model	Temperature; calcium lactate; Day-0 crack width	Day-28 healing efficiency (%)	8 independent conditions
Post-healing strength model	Temperature; calcium lactate; Day-0 crack width	Post-healing compressive strength (MPa)	24 specimens in 8 conditions

Day-28 crack width was not included as an input to the healing-efficiency model. Healing efficiency is calculated directly from Day-0 crack widths; therefore, including the final crack width would provide the model with information that mathematically determines the target. Excluding it allowed the model to operate as an early forecast based on the condition at crack induction.

3.11.5 XGBoost Model Configuration and Training

The models were implemented using XGBoost regression with a squared-error objective. A common low-complexity configuration was used for both targets to reduce the tendency to overfit the small dataset. The model consisted of 100 sequential decision-tree estimators, each limited to a maximum depth of one. The learning rate was 0.05, the minimum child weight was 1, and L2 regularization was applied with a value of 1.0. The subsample and feature-subsample ratios were both set to 1.0, and a fixed random state of 42 was used for reproducibility.

Table-3- 7 XGBoost configuration and validation strategy

Item	Healing-efficiency model	Post-healing strength model
Algorithm	XGBoost regressor	XGBoost regressor
Estimators / maximum depth	100 / 1	100 / 1
Learning rate / L2 regularization	0.05 / 1.0	0.05 / 1.0
Validation	Leave-One-Out	Leave-One-Condition-Out
Held-out unit	One healing sequence	All 3 specimens from one condition

3.11.6 Cross-Validation and Performance Measures

Leave-One-Out cross-validation was used for the healing-efficiency model because only eight independent sequences were available. In each fold, seven records were used for training and the remaining record was predicted. The process was repeated until each condition had served once as the test record.

Leave-One-Condition-Out cross-validation was used for post-healing compressive strength. The condition identification code was used as the grouping variable, and all three strength specimens belonging to one calcium-lactate-temperature condition were held out together. This prevented replicate specimens with identical predictors from appearing simultaneously in the training and test data.

3.11.7 Final Model Fitting, Saving, and Prediction Domain

After cross-validation, each model was fitted to its complete measured dataset and saved in native XGBoost JSON and Joblib formats. The saved models retained the exact feature order of temperature, calcium lactate concentration, and Day-0 crack width.

The valid prediction domain was restricted to the measured ranges: -15°C to 50°C for temperature, 0.5% to 1.5% for calcium lactate concentration, and 0.279 to 0.722 mm for Day-0 crack width. Predictions outside these ranges were treated as unsupported extrapolations. The models were developed as exploratory research tools and were not

intended for structural design, safety assessment, or application to other mortar compositions without external validation.

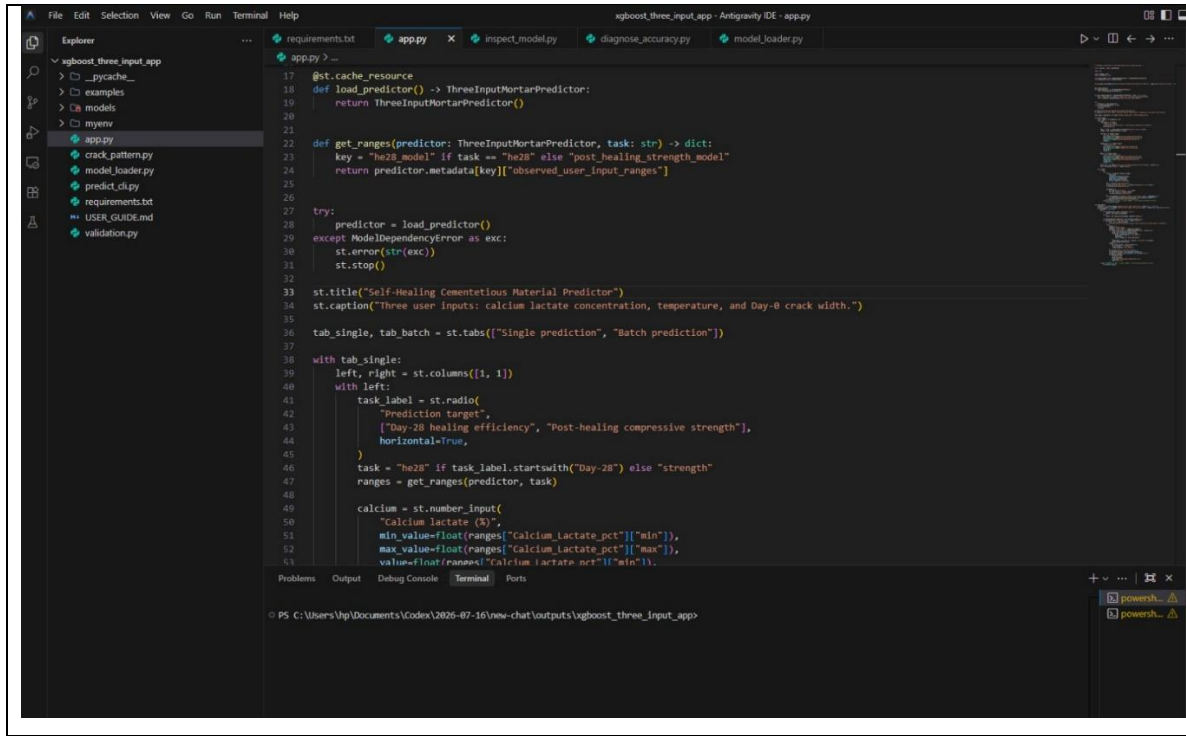


Figure 3- 14 Snapshot of the completed XGBoost model-development and validation workflow.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents and discusses the experimental findings obtained from the pre-crack compressive-strength test, controlled crack induction, ImageJ-based crack monitoring, healing-efficiency calculations, and post-healing compressive-strength testing. Results are reported for the Control mortar and Batches A, B, and C during the pre-crack assessment, while the temperature-controlled healing analysis is limited to Batch B (0.5% calcium lactate) and Batch C (1.5% calcium lactate), in accordance with the experimental program described in Chapter 3.

The findings are organized by test stage so that the effects of calcium lactate dosage and healing temperature can be examined separately and then compared. Numerical results are presented before interpretation. Because the crack-width sequences were obtained from one representative specimen for each batch-temperature combination, those results are treated as descriptive observations rather than group means. Post-healing compressive-strength results, by contrast, are based on three specimens for each batch at each temperature.

4.2 Pre-Crack Compressive Strength

4.2.1 Results

The pre-crack compressive strength was determined after 28 days of water curing. Three specimens were tested for the Control mortar and for each bacterial batch. Table 4-1 presents the individual values, arithmetic mean, and sample standard deviation in MPa. The mean strength was highest for Batch C, followed by Batch B, the Control mortar, and Batch A.

Table - 4 - 1 Pre-crack compressive strength after 28-day curing

Group	Calcium lactate (% by cement mass)	Specimen 1 (MPa)	Specimen 2 (MPa)	Specimen 3 (MPa)	Mean (MPa)	SD (MPa)
Control	0	24.52	26.95	25.62	25.70	1.22
Batch B	0.5	30.67	32.31	29.95	30.98	1.21
Batch C	1.5	33.45	32.31	30.79	32.18	1.33
Batch A	10	12.92	11.78	12.35	12.35	0.57

As shown in Figure 4-1, the error bars represent the sample standard deviation of the three measurements in each group. The figure is provided to support visual comparison of the group means and does not imply inferential statistical significance.

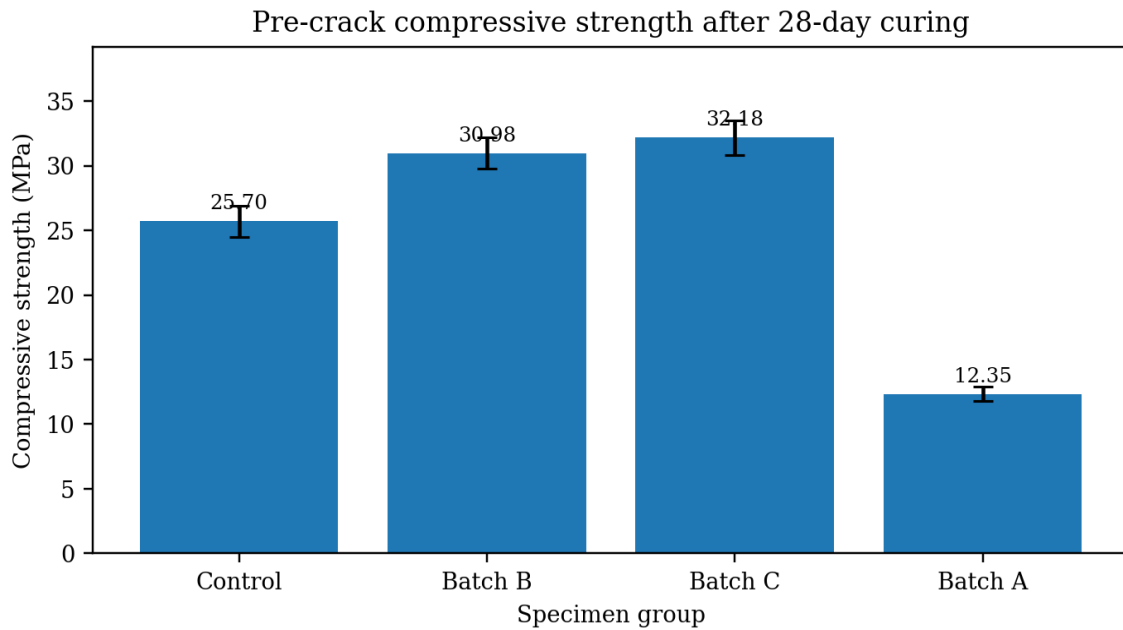


Figure 4 - 1 Mean pre-crack compressive strength with sample standard-deviation error bars.

4.2.2 Discussion

Batch C produced the highest mean pre-crack compressive strength of 32.18 MPa, while Batch B produced 30.98 MPa. These values were 25.2% and 20.5% higher than the Control mean of 25.70 MPa, respectively. The observed increase at the lower calcium lactate dosages is compatible with published findings that moderate calcium lactate incorporation in bacterial cementitious materials can improve matrix densification and compressive strength [16]. Microbial carbonate precipitation has also been associated with deposition within pores and discontinuities, which can contribute to a denser cementitious matrix [7], [58]. Mineralogical characterization was not undertaken in the present study and the mechanism discussed here is instead an explanation based on literature that might be expected to be present in the formation of calcium carbonate in these specimens.

The mean strength of Batch A (10% calcium lactate) was 12.35 MPa, 51.9% lower than the mean strength of the Control. The results as such, therefore, indicated a non-monotonic dosage response with an improvement in strength at 0.5% and 1.5% and a drastic reduction at 10%. Too high a nutrient addition can upset the matrix hydration, change the apparent solid components, or introduce discontinuities into the matrix, while controlled doses can create a more favorable relationship between bacterial activity and cementitious integrity [16]. This interpretation should be made with caution as Batch A was granular calcium lactate whereas Batches B & C were as a powder. Thus, the problem of dosage and physical form were not entirely independent variables.

4.3 Crack Induction and Selection of Specimens for the Healing Study

4.3.1 Crack-Induction Observations

Using the controlled wire assisted loading as described in Chapter 3, the specimens were loaded until the first surface crack was observed and the loading was then stopped. The Day 0 widths of the representative specimens from Batches B and C varied from 0.279 to 0.722 mm. These differences were preserved as the initial condition in the image-based analysis and the healing efficiency was later normalized by the width of each specimen at Day 0. A qualitative preliminary observation was also taken for Batch A. The opened crack induced by it seemed to be closed quickly within about 24 h, but this was not included in the quantitative 4-temperature analysis. The supporting Day 0 and ~24 h photographs are

kept on Figure 4-2. The photographs should be used as qualitative evidence as no corresponding numerical series of the crack-width was provided for Batch A.



Figure 4 - 2 Photographic comparison of the Batch A crack at Day 0 and approximately 24 h after induction.

4.3.2 Rationale for Excluding Batch A

Batch A was excluded from the temperature-controlled healing phase for two experimentally documented reasons. First, it exhibited the lowest pre-crack compressive strength and therefore provided a weaker mechanical basis for comparison with Batches B and C. Second, the preliminary visual observation indicated very rapid crack closure, which would have provided limited resolution for the planned Day 7, 14, 21, and 28 monitoring schedule. The formal incubation and comparative healing study was therefore restricted to Batches B and C. No post-healing Batch A results are reported in this chapter.

4.4 Crack-Healing Performance of Batches B and C

4.4.1 Crack-Width Reduction Over Time

Crack width was measured at the same ImageJ measurement point on Day 0, Day 7, Day 14, Day 21, and Day 28. One representative specimen was selected from each batch at each temperature. Table 4-2 therefore presents specimen-specific crack-width sequences rather than the means of three replicates. This distinction is important when interpreting the photographic and line-graph trends.

Table - 4 - 2 Representative crack-width measurements during the 28-day healing period

Temp. (°C)	Batch	Calcium lactate (%)	Day 0 (mm)	Day 7 (mm)	Day 14 (mm)	Day 21 (mm)	Day 28 (mm)
-15	Batch B	0.5	0.369	0.369	0.347	0.298	0.184
-15	Batch C	1.5	0.448	0.448	0.421	0.362	0.224
20	Batch B	0.5	0.722	0.649	0.469	0.173	0.000
20	Batch C	1.5	0.395	0.343	0.161	0.067	0.000
40	Batch B	0.5	0.413	0.338	0.181	0.041	0.000
40	Batch C	1.5	0.590	0.466	0.142	0.000	0.000
50	Batch B	0.5	0.279	0.245	0.133	0.053	0.000
50	Batch C	1.5	0.496	0.406	0.183	0.059	0.000

At -15°C , neither representative specimen showed measurable closure by Day 7, and both retained approximately half of their original crack width on Day 28. At 20°C , 40°C , and 50°C , the measured crack width reached 0.000 mm by Day 28 for both batches. The fastest recorded sequence occurred for Batch C at 40°C , where the fixed measurement point reached 0.000 mm by Day 21. Figure 4-3 presents the corresponding crack-width trends.

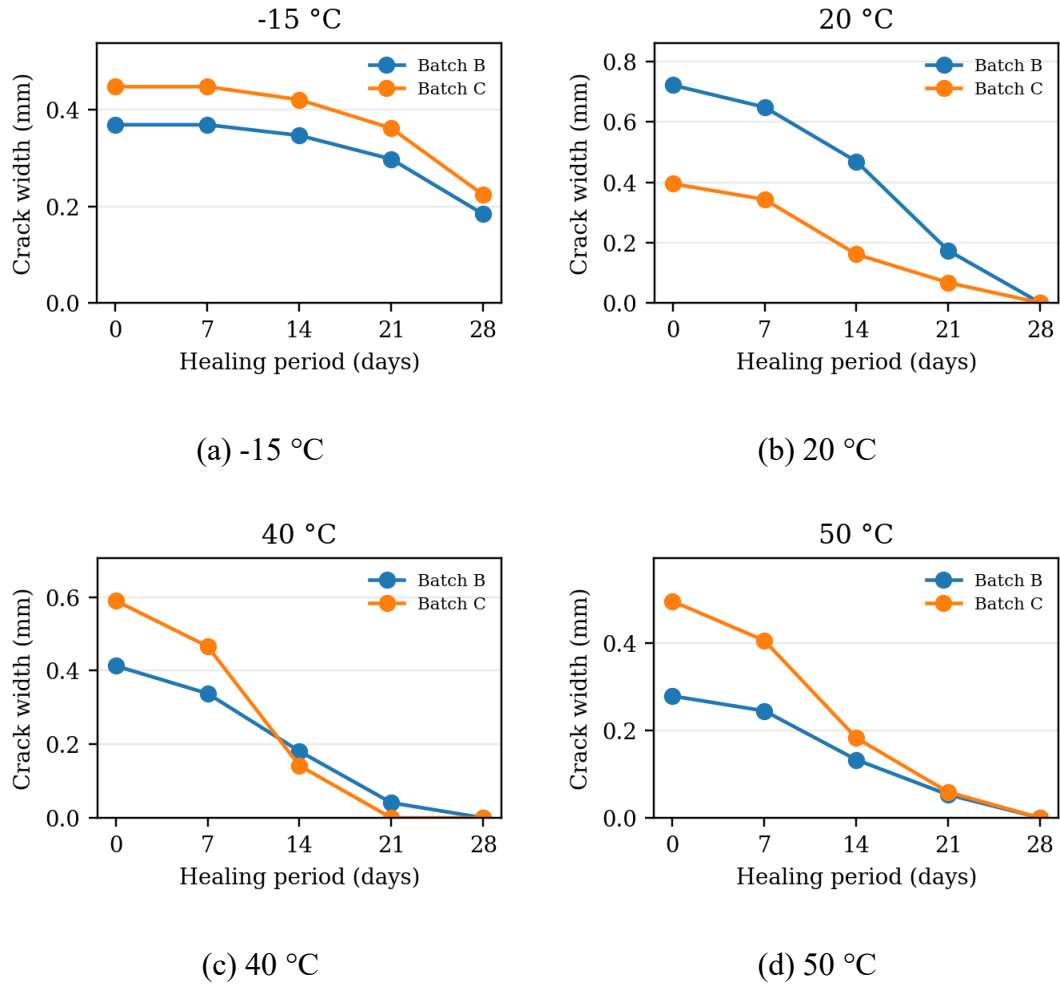


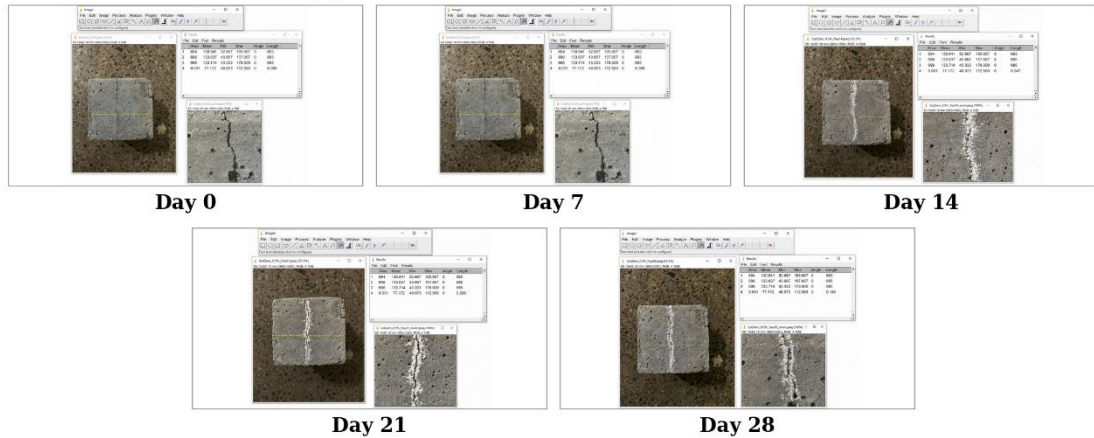
Figure 4 - 3 Representative crack-width reduction for Batches B and C at the four healing temperatures.

4.4.2 Photographic Evidence of Healing

Photographic evidence should relate directly with the specimens and fixed measurement points listed in Table 4-2. A total of 40 figures is not given, but four composite figures, one for each temperature. There are 5 sets of the sequence Day 0, 7, 14, 21, 28 for each one of the selected Batch B and Batch C specimens. The complete photographic archive may be retained in an appendix or laboratory record, while the quantitative tables and graphs remain the principal evidence.

Figure 4-4 presents the representative ImageJ photographic sequence at -15°C . The same selected specimen and fixed crack location are shown at each observation day.

Healing temperature: -15°C
(a) Batch B - 0.5% calcium lactate



(b) Batch C - 1.5% calcium lactate

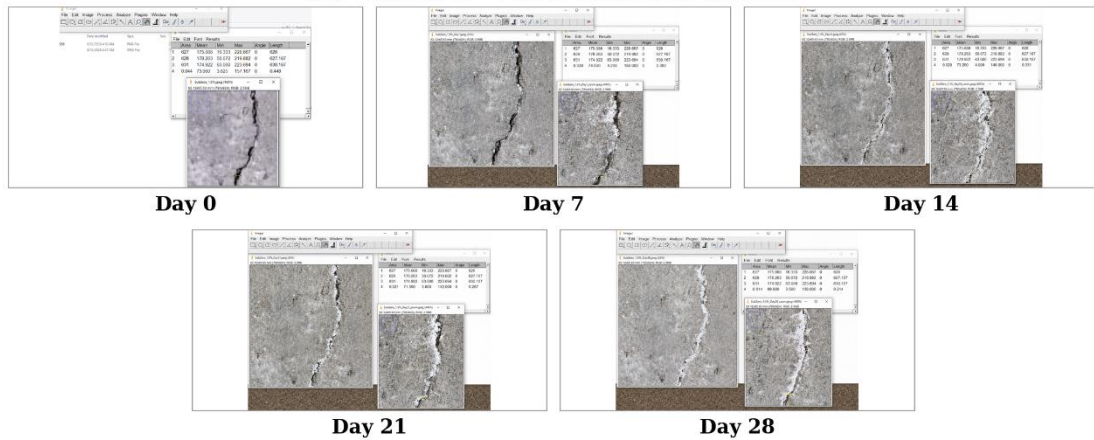
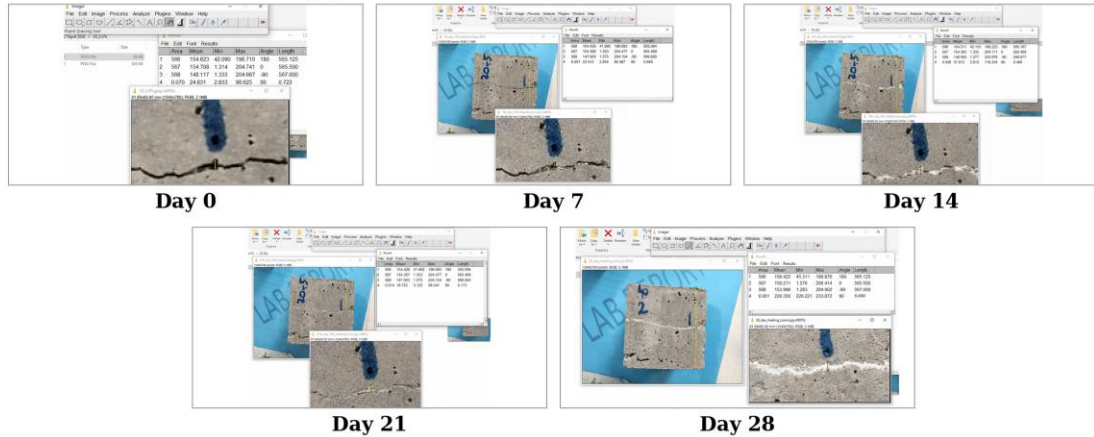


Figure 4 - 4 Representative ImageJ crack-monitoring sequences for Batch B (0.5% calcium lactate) and Batch C (1.5% calcium lactate) at -15°C from Day 0 to Day 28.

Figure 4-5 presents the representative ImageJ photographic sequence at 20 °C. The same selected specimen and fixed crack location are shown at each observation day.

Healing temperature: 20 °C
(a) Batch B - 0.5% calcium lactate



(b) Batch C - 1.5% calcium lactate

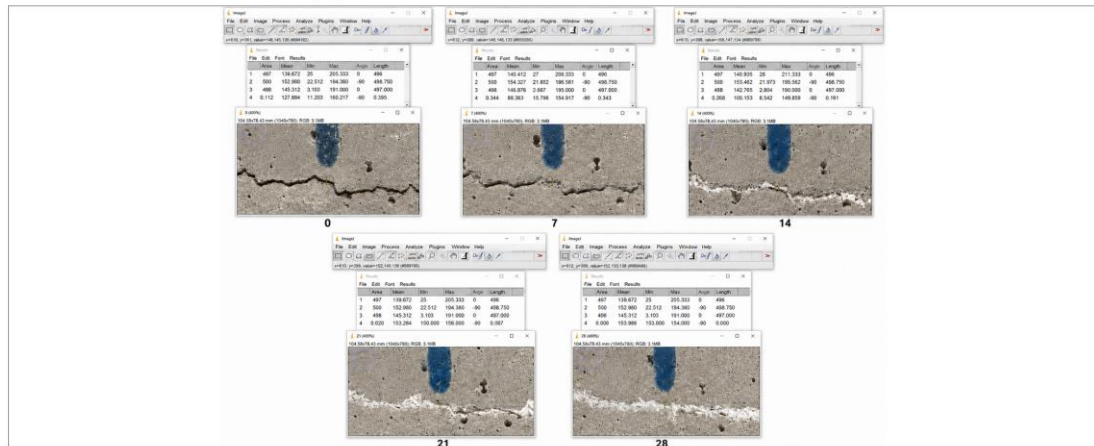
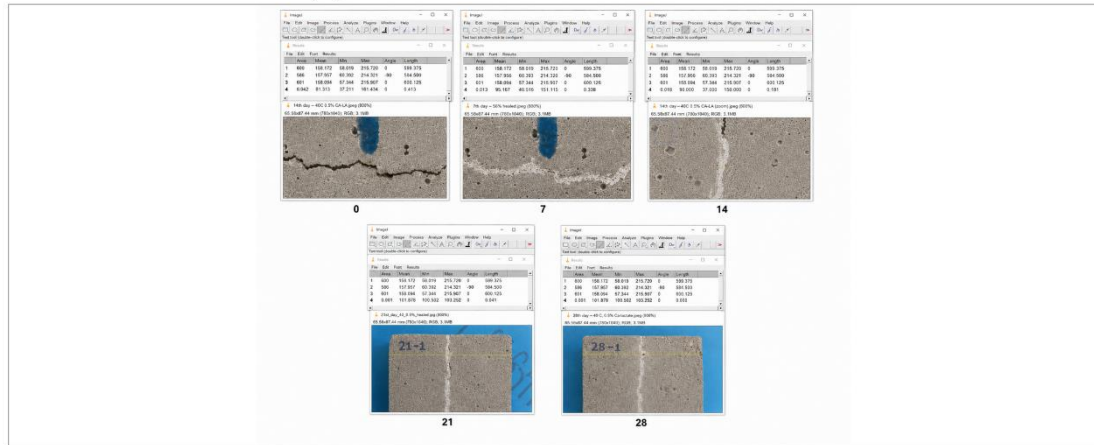


Figure 4 - 5 Representative ImageJ crack-monitoring sequences for Batch B (0.5% calcium lactate) and Batch C (1.5% calcium lactate) at 20 °C from Day 0 to Day 28.

Figure 4-6 presents the representative ImageJ photographic sequence at 40 °C. The same selected specimen and fixed crack location are shown at each observation day.

Healing temperature: 40 °C
(a) Batch B - 0.5% calcium lactate



(b) Batch C - 1.5% calcium lactate

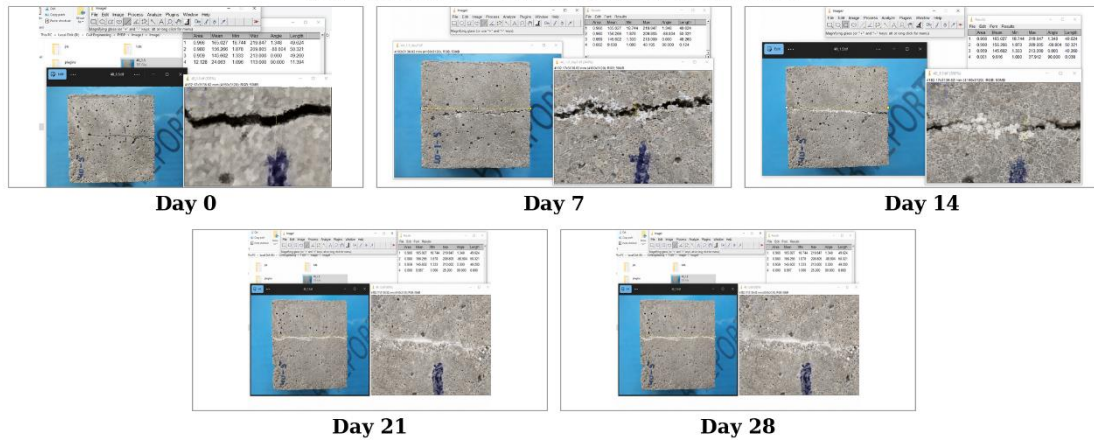
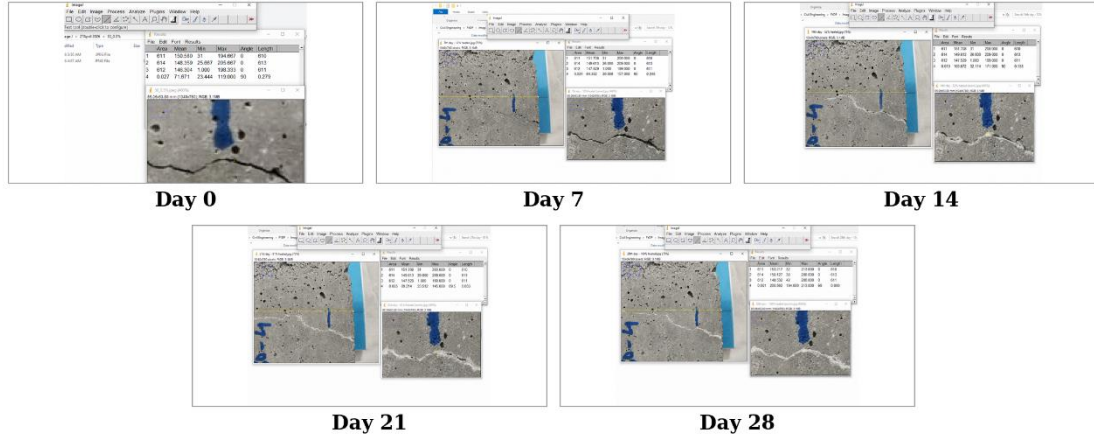


Figure 4 - 6 Representative ImageJ crack-monitoring sequences for Batch B (0.5% calcium lactate) and Batch C (1.5% calcium lactate) at 40 °C from Day 0 to Day 28.

Figure 4-7 presents the representative ImageJ photographic sequence at 50 °C. The same selected specimen and fixed crack location are shown at each observation day.

Healing temperature: 50 °C

(a) Batch B - 0.5% calcium lactate



(b) Batch C - 1.5% calcium lactate

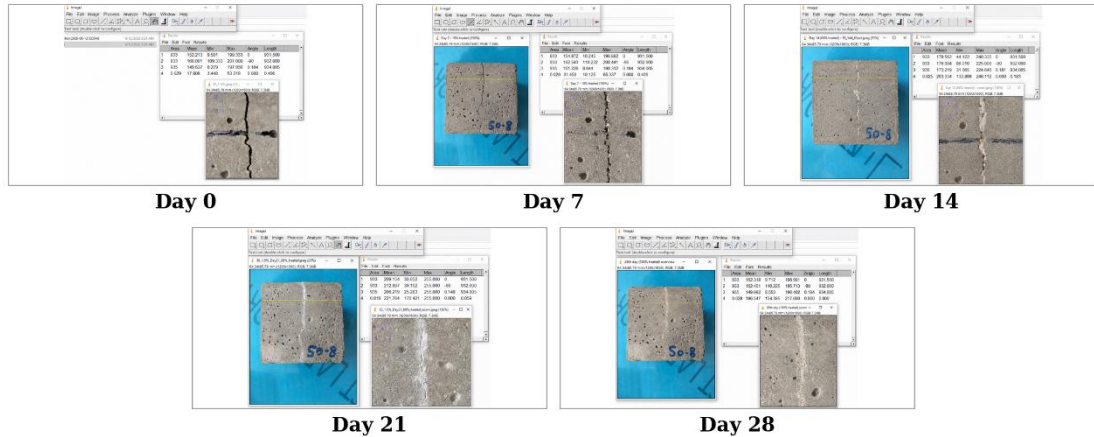


Figure 4 - 7 Representative ImageJ crack-monitoring sequences for Batch B (0.5% calcium lactate) and Batch C (1.5% calcium lactate) at 50 °C from Day 0 to Day 28.

4.4.3 Discussion

The temperature-dependent pattern was clear at the representative measurement points. Healing proceeded slowly at -15°C , where both batches showed no measurable change during the first seven days and only approximately 50% closure after 28 days. In contrast, the specimens exposed to 20°C , 40°C , and 50°C reached complete apparent surface closure at the fixed point by Day 28. Bacterial self-healing depends on the viability and metabolic activity of the microorganisms as well as the availability of moisture and nutrients; environmental conditions can therefore strongly influence mineral precipitation and crack repair [6], [63]. The sub-zero response in the present study is consistent with suppression or substantial slowing of biological and chemical healing processes under cold exposure.

The warmest of the conditions was 40°C which resulted in the quickest overall response. Both Batch C and Batch B were completely closed by Day 21, and Batch B was 90.1% closed at the same age. Both batches closed completely at 50°C , although the intermediate healing rates were less than at 40°C . This indicates that the optimum within the range tested was not the highest temperature observed but rather it was 40°C that gave the best observed balance for the selected specimens. Generally at the same temperature, Batch C healed faster than Batch B, especially at 20°C and 40°C , meaning that it required a stronger dosage (1.5%) than 0.5%.

The results obtained from the visual and ImageJ images show closure of the surface cracks, but they do not necessarily indicate the material that has been deposited or indicate full crack filling of the material. The closure of cracks by bacterial activity is believed to be due to mineral precipitation, usually calcium carbonate [6, 7, 18]. Since SEM, XRD, or chemical analysis was not done in this study, the term apparent surface closure is used as appropriate, and the post-healing compressive-strength results are treated separately as proof of mechanical recovery.

4.5 Post-Healing Compressive Strength

4.5.1 Results

After the 28-day healing period, three specimens from Batch B and three specimens from Batch C were tested at each temperature. Table 4-3 presents the individual results, mean, sample standard deviation, and strength recovery relative to the corresponding pre-crack group mean. Batch A and the Control mortar were not included in this phase.

Table - 4 - 3 Post-healing compressive strength and strength recovery after 28 days

Temp. (°C)	Batch	Calcium lactate (%)	Specimen 1 (MPa)	Specimen 2 (MPa)	Specimen 3 (MPa)	Mean (MPa)	SD (MPa)	Strength recovery (%)
-15	Batch B	0.5	16.91	17.34	17.56	17.27	0.33	55.8
-15	Batch C	1.5	18.82	19.38	19.49	19.23	0.36	59.8
20	Batch B	0.5	21.65	21.25	21.92	21.61	0.34	69.8
20	Batch C	1.5	23.78	23.42	23.96	23.72	0.27	73.7
40	Batch B	0.5	24.74	25.13	24.34	24.74	0.40	79.9
40	Batch C	1.5	27.64	27.32	26.87	27.28	0.39	84.8
50	Batch B	0.5	23.19	22.78	23.41	23.13	0.32	74.7
50	Batch C	1.5	25.22	25.01	24.92	25.05	0.15	77.8

The highest post-healing mean was 27.28 MPa for Batch C at 40 °C, corresponding to 84.8% strength recovery. Batch B also achieved its highest result at 40 °C, with a mean of 24.74 MPa and 79.9% recovery. The lowest means for both batches occurred at −15 °C. Figures 4-8 and 4-9 compare the post-healing strengths and calculated recovery percentages.

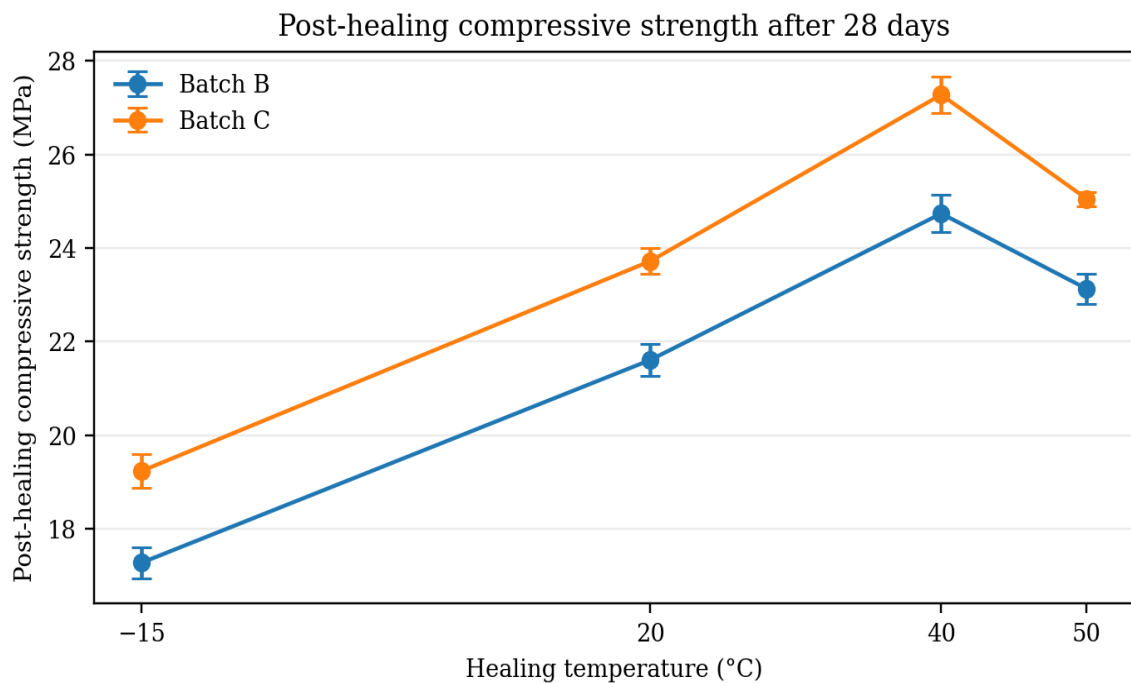


Figure 4 - 8 Mean post-healing compressive strength with sample standard-deviation error bars.

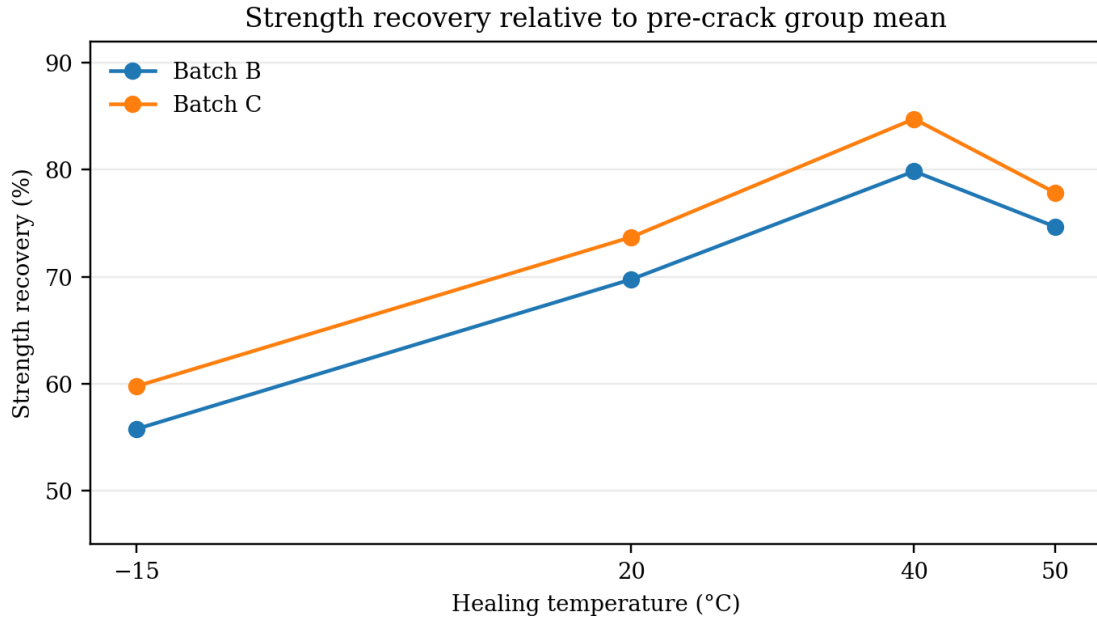


Figure 4 - 9 Strength recovery relative to the corresponding pre-crack compressive-strength mean.

4.5.2 Discussion

The mechanical results followed the main temperature trend observed in the crack-monitoring data. Both batches showed their lowest strength recovery at $-15\text{ }^{\circ}\text{C}$ and their highest recovery at $40\text{ }^{\circ}\text{C}$. At $50\text{ }^{\circ}\text{C}$, strength recovery remained higher than at $20\text{ }^{\circ}\text{C}$ but declined relative to $40\text{ }^{\circ}\text{C}$. The result indicates that the apparent surface closure observed at Day 28 did not correspond to identical mechanical restoration: although both batches reached 100% crack closure at the fixed point at $20\text{ }^{\circ}\text{C}$, $40\text{ }^{\circ}\text{C}$, and $50\text{ }^{\circ}\text{C}$, their strength-recovery values remained different.

This is not surprising since the distinction is based on a measure of surface width and the healed zone's depth, continuity, and internal quality cannot be determined by a two-dimensional surface-width measurement. Surface closure can take place without complete internal repair and the precipitated material may not completely rebuild the original load bearing matrix. The post-healing strength results then complement the measurements taken with ImageJ; therefore they do not duplicate it. The post-healing strength of Batch C was

higher than Batch B at all temperatures, the highest being at 40 °C. The overall evidence therefore suggests that Batch C (40 °C) is the optimum tested condition.

4.6 Healing Efficiency

4.6.1 Calculation and Results

Healing efficiency was calculated from the representative crack-width sequence using the relationship given in Equation (4-1).

$$\text{Healing Efficiency (\%)} = \frac{W_0 - W_t}{W_0} \times 100 \quad \text{Equation 4-1}$$

In Equation (4-1), w_0 is the Day 0 crack width and w_t is the crack width at the selected observation day. Table 4-4 consolidates the calculated values. The corrected Batch B efficiencies at 20 °C are 10.1%, 35.0%, 76.0%, and 100.0% on Days 7, 14, 21, and 28, respectively.

Table - 4 - 4 : Healing-efficiency summary for representative specimens

Temp. (°C)	Batch	Calcium lactate (%)	Day 0 (%)	Day 7 (%)	Day 14 (%)	Day 21 (%)	Day 28 (%)
-15	Batch B	0.5	0.0	0.0	6.0	19.2	50.1
-15	Batch C	1.5	0.0	0.0	6.0	19.2	50.0
20	Batch B	0.5	0.0	10.1	35.0	76.0	100.0
20	Batch C	1.5	0.0	13.2	59.2	83.0	100.0
40	Batch B	0.5	0.0	18.2	56.2	90.1	100.0
40	Batch C	1.5	0.0	21.0	75.9	100.0	100.0
50	Batch B	0.5	0.0	12.2	52.3	81.0	100.0
50	Batch C	1.5	0.0	18.1	63.1	88.1	100.0

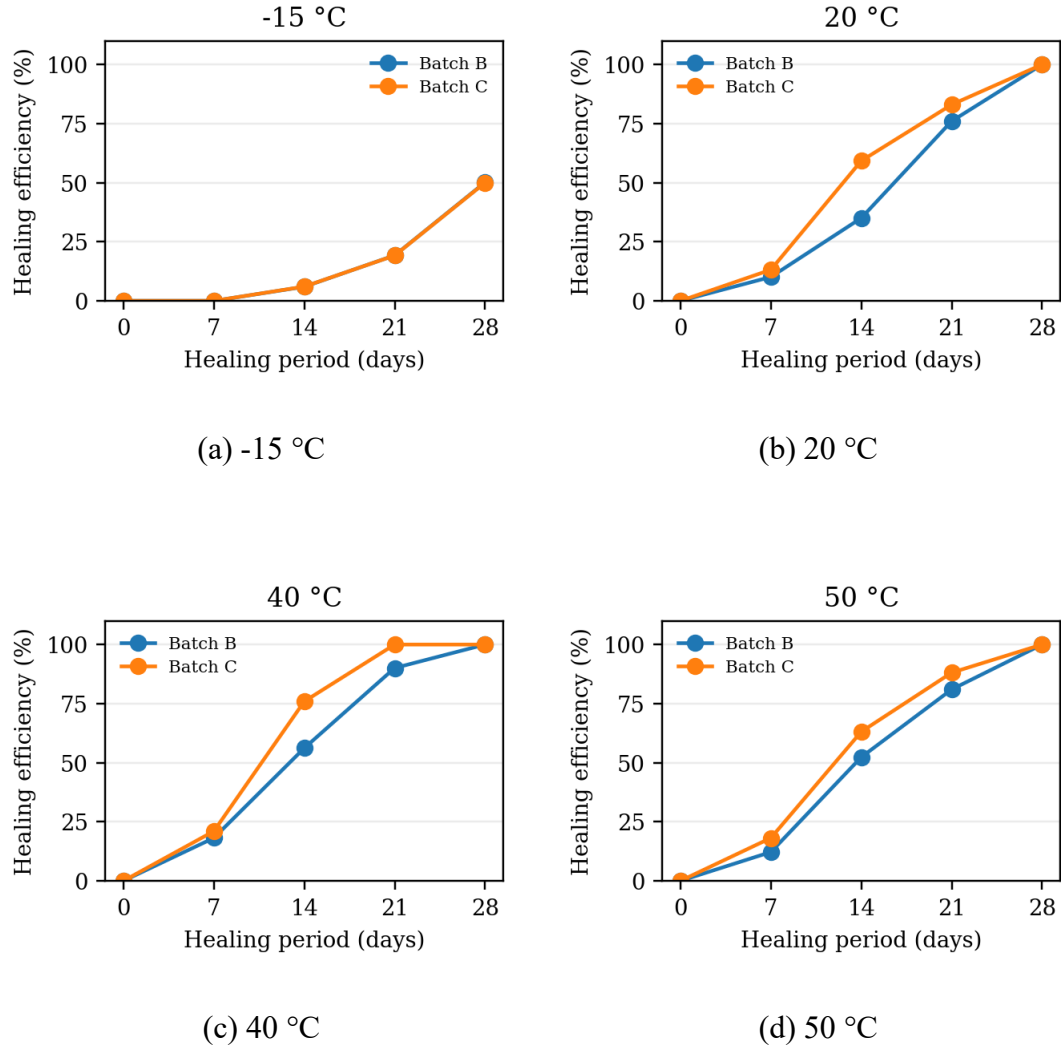


Figure 4 - 10 Healing-efficiency development for the representative specimens at each temperature.

4.6.2 Best-Performing Combination

At Day 28, both batches achieved 100% apparent crack closure at 20 °C, 40 °C, and 50 °C; therefore, final surface closure alone did not identify a unique optimum. The best overall condition was selected by considering the rate of crack closure together with post-healing mechanical recovery. Batch C at 40 °C reached 100% closure by Day 21 and achieved the highest post-healing mean of 27.28 MPa and the highest strength recovery of 84.8%. Accordingly, within the investigated conditions, Batch C containing 1.5% calcium lactate at 40 °C was identified as the best-performing combination.

4.7 Comparative Analysis

4.7.1 Effect of calcium lactate Concentration

The pre-crack results indicate that the calcium lactate dosage had no linear response. Batch B (0.5%) and Batch C (1.5%) were more powerful than the Control, while Batch A (10%) was significantly less powerful. For intermediate crack closure and recovery in post-healing strength Batch C generally exhibited a superior performance compared to Batch B, within the healing period. These results showed that 1.5% worked better than 0.5% for the investigated bacterial mortar and 10% was too much for the pre-crack mechanical response.

The comparison does not provide a standard optimum dose. Only three doses were tested and Batch A was not tested at all on the different temperatures, besides its calcium lactate was granular instead of powdered. The observed trend should therefore be described, as an experiment optimum, for the tested materials and procedures, and not as a general design recommendation.

4.7.2 Effect of Temperature

Temperature exerted a strong descriptive influence on both crack closure and mechanical recovery. The -15°C condition delayed crack closure and produced the lowest post-healing strength for both batches. Increasing the temperature to 20°C accelerated closure, while 40°C produced the fastest representative healing sequence and the highest strength recovery. At 50°C , complete surface closure was still recorded by Day 28, but the intermediate rates and mechanical recovery were lower than at 40°C . The tested response therefore followed an increase from sub-zero to 40°C followed by a modest decline at 50°C .

4.8 Comparison with Previous Studies

The pre-crack strength trend is broadly consistent with previous work showing that calcium lactate dosage requires optimization. Vijay and Murmu reported a maximum compressive-strength increase of 12% at 0.5% calcium lactate in their bacterial concrete system [16]. In the present mortar, Batch B at 0.5% and Batch C at 1.5% exceeded the Control mean by 20.5% and 25.2%, respectively, whereas the 10% Batch A mixture was markedly weaker. Achal, Mukherjee, and Reddy reported a 36% compressive-strength increase for bacteria-

treated mortar [58]. The present increases were lower than that value, but direct numerical equivalence is not expected because bacterial species preparation, concentration, nutrient dosage, mix proportions, specimen size, and curing conditions differ among studies.

Wiktor and Jonkers reported healing of cracks up to 0.46 mm in bacteria-based concrete after prolonged water exposure [18]. The representative initial widths in the present study ranged from 0.279 to 0.722 mm, and complete apparent closure at the fixed surface point was recorded by Day 28 at 20 °C, 40 °C, and 50 °C. This comparison supports the feasibility of bacterial crack closure but should be interpreted cautiously: the present values were obtained from one representative specimen per batch-temperature combination, the specimens were maintained at high relative humidity rather than immersed in water, and the internal extent and mineral composition of the healed region were not characterized.

This lack of response at −15 °C and the differences in the warmer exposures are similar to the overall trend that the crack width and environmental exposure affect the performance of bacteria-based self-healing systems [63]. The present study extends that general observation by directly comparing four temperatures under one mortar *program*, while also demonstrating that complete surface closure does not necessarily imply complete compressive-strength restoration.

4.9 Machine-Learning Prediction

This section reports the completed preliminary machine-learning component. Two XGBoost regression models were developed from the measured experimental records: one for Day-28 healing efficiency and one for post-healing compressive strength. The models are interpreted as bounded proof-of-concept tools because the dataset was small, clustered and limited to the tested material system.

4.9.1 Completed Model Configuration and Validation

Subsequent to the initial preparation of this chapter, the machine-learning component was completed. Two XGBoost regression models were trained using temperature, calcium lactate concentration, and Day-0 crack width. The first model predicted Day-28 healing efficiency, and the second predicted post-healing compressive strength.

The healing-efficiency model was evaluated using Leave-One-Out cross-validation across eight independent conditions. The strength model was evaluated using Leave-One-Condition-Out cross-validation across eight batch-temperature groups, with the three replicate specimens from each group held out together. The cross-validated performance is summarized in Table 4-5.

4.9.2 Prediction of Day-28 Healing Efficiency

The Day-28 healing-efficiency model produced a cross-validated MAE of 1.00 percentage points and an RMSE of 1.71 percentage points. The corresponding R^2 was 0.994, and the mean bias was 0.70 percentage points. Table 4-5 compares the measured and cross-validated predicted values.

The apparently strong fit must be interpreted cautiously. Six of the eight experimental conditions recorded 100% apparent surface closure on Day 28, while the two $-15\text{ }^{\circ}\text{C}$ conditions recorded approximately 50% closure. The model therefore learned a highly separated temperature pattern rather than a continuous response across a broad range of healing efficiencies. The result demonstrates that the tested sub-zero condition could be distinguished from the warmer tested conditions, but it does not establish general predictive accuracy for intermediate healing responses.

4.9.3 Prediction of Post-Healing Compressive Strength

The post-healing compressive-strength model produced a grouped cross-validated MAE of 2.18 MPa and an RMSE of 2.57 MPa. The R^2 value was 0.298, while the mean bias was 0.22 MPa. Figure 4-5 shows the measured and Leave-One-Condition-Out predicted specimen strengths.

The strength model showed limited-to-modest grouped cross-validated performance and did not reproduce all condition-level differences. This outcome is consistent with the limited information available from three initial-condition predictors. Post-healing compressive strength depends not only on temperature, nutrient dosage and initial surface crack width, but also on internal crack geometry, specimen-to-specimen variability, matrix

damage, extent of mineral deposition and the quality of internal healing. These factors were not represented directly in the model inputs.

4.9.4 Condition-Level Predictions and Feature Importance

Table 4-6 presents the measured and cross-validated predictions for the eight experimental conditions. The strength values are shown as condition means to permit direct comparison with the condition-level healing-efficiency records.

Table - 4 - 5 Condition-level measured and cross-validated predictions

Temp (°C)	CaL (%)	Day-0 width (mm)	Measured HE28 (%)	Predicted HE28 (%)	Measured strength (MPa)	Predicted strength (MPa)
-15	0.5	0.369	50.1	72.10	17.27	19.00
-15	1.5	0.448	50.0	72.15	19.23	19.29
20	0.5	0.722	100.0	93.24	21.61	21.7
20	1.5	0.395	100.0	95.60	23.72	23.69
40	0.5	0.413	100.0	95.60	24.74	24.70
40	1.5	0.590	100.0	95.60	27.28	25.66
50	0.5	0.279	100.0	95.60	23.13	23.12
50	1.5	0.496	100.0	94.29	25.05	24.95

4.9.5 Practical Interpretation and Limitations

The significant practical benefit of the completed models is that they will only use variables that can be known at the start of the healing period. Selected treatment condition is described by temperature and calcium lactate concentration, and initial visible damage is described by Day-0 crack width. Therefore, the healing-efficiency model could potentially be used to produce an early Day-28 prediction (before a final crack measurement) while the strength model could be used for an early mechanical prediction (before destructive post-healing testing).

The predictions are confined to the range of measured domain of (0.5% to 1.5%) calcium lactate, (-15°C to 50°C) and Day-0 crack widths of (0.279 to 0.722). Not all conditions can be predicted by the models, such as 0.2% of calcium lactate or 60°C , as XGBoost models of trees will not give reliable results when predicting outside the splits they were trained to learn.

The healing-efficiency model utilized only eight independent records, and the target values were tightly grouped. Strength model only required 24 measurements on specimens and 8 distinct sets of inputs. The models produced are therefore proof of concept models and need to be validated externally with further independently monitored samples, more calcium lactate concentrations, more temperatures and a more extensive range of initial crack widths.

Self-Healing Cementitious Material Predictor
Three user inputs: calcium lactate concentration, temperature, and Day-0 crack width.

Single prediction Batch prediction

Prediction target
☒ Day-28 healing efficiency ☐ Post healing compressive strength

Calcium lactate (%)
1.50

Temperature (C)
40.00

Day-0 crack width (mm)
0.590

☐ Allow out of range predictions with warning

Predict

Predicted value
95.60 %

Within observed training range.

Estimated internal crack width inputs

Day7_Crack_Width_mm	Day14_Crack_Width_mm	Day21_Crack_Width_mm
0	0.466	0.342

Later crack widths were estimated from observed training pattern trajectories; they were not supplied by the user.

Self-Healing Cementitious Material Predictor
Three user inputs: calcium lactate concentration, temperature, and Day-0 crack width.

Single prediction Batch prediction

Prediction target
☐ Day-28 healing efficiency ☒ Post healing compressive strength

Calcium lactate (%)
1.50

Temperature (C)
40.00

Day-0 crack width (mm)
0.590

☐ Allow out of range predictions with warning

Predict

Predicted value
25.66 MPa

Within observed training range.

Estimated internal crack width inputs

Later crack widths were estimated from observed training pattern trajectories; they were not supplied by the user.

Figure 4 - 11 Snapshots of the completed machine-learning prediction interface.

4.10 Summary of Key Findings

The pre-crack results showed the highest mean compressive strength for Batch C (1.5% calcium lactate), followed by Batch B (0.5%), the Control mortar and Batch A (10%).

During healing, the representative specimens at 20 °C, 40 °C and 50 °C reached complete apparent surface closure by Day 28, whereas the −15 °C specimens achieved approximately 50% closure. Batch C at 40 °C was the best-performing tested combination because it reached complete closure by Day 21 and produced the highest post-healing strength recovery of 84.8%. The results also indicated that surface crack closure and mechanical recovery were related but not equivalent, providing the basis for the recommendations presented in Chapter 5.

The completed machine-learning component further indicated that temperature, calcium lactate dosage and Day-0 crack width can be used as initial-condition inputs for preliminary prediction. The healing-efficiency model distinguished the sub-zero and warmer Day-28 outcomes, while the post-healing strength model achieved an MAE of 2.18 MPa under leave-one-condition-out validation. These results support the feasibility of data-driven prediction but also indicate the need for a larger and more varied experimental dataset.

CHAPTER 5

RECOMMENDATIONS

5.1 Recommendations for Experimental Application

The following recommendations are based on the experimental and machine-learning findings reported in this thesis and are limited to the materials, specimen size, mixture proportions, healing conditions, and measured ranges investigated in the study.

1. For further laboratory verification using the same mortar system, Batch C containing 1.5% calcium lactate at 40 °C should be treated as the best-performing tested condition because it produced the fastest complete apparent surface closure and the highest post-healing compressive-strength recovery. This result should not be presented as a universal optimum without additional validation.
2. A calcium lactate dosage of 10% should not be adopted in the same mortar formulation without reformulation and further testing because Batch A produced the lowest pre-crack compressive strength among the investigated groups.
3. Moisture availability should be maintained during the healing period because the temperature-controlled specimens were evaluated under approximately 95% relative humidity. The reported healing response should not be assumed to represent dry-service conditions.
4. Surface crack closure and post-healing compressive strength should be evaluated together. Complete apparent closure at the measured surface point did not produce identical strength recovery at 20 °C, 40 °C, and 50 °C.
5. The completed XGBoost models should be used only as preliminary screening tools within the measured ranges of –15 °C to 50 °C, 0.5% to 1.5% calcium lactate, and 0.279 to 0.722 mm Day-0 crack width. They should not be used for structural design or unsupported extrapolation.

5.2 Recommendations for Further Research

1. Future crack-monitoring *programs* should use multiple quantitatively measured specimens for every batch temperature condition and should report the specimen-selection procedure where representative images are presented.
2. There should be non-bacterial control specimens that are subjected to the same temperature to allow for the recognition of bacterial healing from autogenous healing, hydration, carbonation and moisture effects.
3. The additional calcium lactate dosages (between 0.5 and 1.5) and additional temperatures (around 40 °C) should be investigated to define the response more precisely. Additional initial crack widths should also be taken into account, to get a more balanced and wider range of initial widths.
4. If possible, the composition, the distribution of composition, the internal continuity, and the functional effectiveness of the healing products should be verified by performing SEM, EDS, XRD, water-absorption, permeability tests, etc.
5. Viability, spore survival, available nutrients, relative humidity control, and long term performance should be documented more extensively and then larger specimens and exposure conditions that reflect the field.
6. New observations, including independently monitored data and unique inputs per specimen, should be added to the datasets for the machine learning. Future modelling efforts should benchmark XGBoost with simple baseline regression models, and alternative regression models; assess prediction uncertainty; and validate prediction using external data prior to application.

REFERENCE LIST

- [1] R. M. Andrew, "Global CO₂ emissions from cement production," *Earth System Science Data*, vol. 10, no. 1, pp. 195–217, 2018, doi: 10.5194/essd-10-195-2018.
- [2] X.-Y. Wang, "Effects of crack and climate change on service life of concrete subjected to carbonation," *Applied Sciences*, vol. 8, no. 4, Art. no. 572, 2018, doi: 10.3390/app8040572.
- [3] K. Vijay, M. Murmu, and S. V. Deo, "Bacteria based self healing concrete—A review," *Construction and Building Materials*, vol. 152, pp. 1008–1014, 2017, doi: 10.1016/j.conbuildmat.2017.07.040.
- [4] S. Gupta, S. D. Pang, and H. W. Kua, "Autonomous healing in concrete by bio-based healing agents—A review," *Construction and Building Materials*, vol. 146, pp. 419–428, 2017, doi: 10.1016/j.conbuildmat.2017.04.111.
- [5] P. Y. Wong, J. Mal, A. Sandak, L. Luo, J. Jian, and N. Pradhan, "Advances in microbial self-healing concrete: A critical review of mechanisms, developments, and future directions," *Science of the Total Environment*, vol. 947, Art. no. 174553, 2024, doi: 10.1016/j.scitotenv.2024.174553.
- [6] H. M. Jonkers, A. Thijssen, G. Muyzer, O. Copuroglu, and E. Schlangen, "Application of bacteria as self-healing agent for the development of sustainable concrete," *Ecological Engineering*, vol. 36, no. 2, pp. 230–235, 2010, doi: 10.1016/j.ecoleng.2008.12.036.
- [7] W. De Muynck, N. De Belie, and W. Verstraete, "Microbial carbonate precipitation in construction materials: A review," *Ecological Engineering*, vol. 36, no. 2, pp. 118–136, 2010, doi: 10.1016/j.ecoleng.2009.02.006.
- [8] J. Wang, K. Van Tittelboom, N. De Belie, and W. Verstraete, "Use of silica gel or polyurethane immobilized bacteria for self-healing concrete," *Construction and Building Materials*, vol. 26, no. 1, pp. 532–540, 2012, doi: 10.1016/j.conbuildmat.2011.06.054.
- [9] N. Yamasmit, P. Sangkeaw, W. Jitchaijaroen, C. Thongchom, S. Keawsawasvong, and V. Kamchoom, "Effect of *Bacillus subtilis* on mechanical and self-healing properties in mortar with different crack widths and curing conditions," *Scientific Reports*, vol. 13, Art. no. 7844, 2023, doi: 10.1038/s41598-023-34837-x.
- [10] R. Z. Rahmawan, Erizal, H. Putra, P. G. Oktafiani, and Sutoyo, "Effectiveness of concrete crack repair using *Bacillus subtilis* and calcium lactate," *INERSIA*, vol. 21, no. 1, pp. 125–134, 2025, doi: 10.21831/inersia.v21i1.71313.
- [11] W. De Muynck, K. Cox, N. De Belie, and W. Verstraete, "Bacterial carbonate precipitation as an alternative surface treatment for concrete," *Construction and Building Materials*, vol. 22, no. 5, pp. 875–885, 2008, doi: 10.1016/j.conbuildmat.2006.12.011.
- [12] I. M. Elgendy, N. E. Elkaliny, H. M. Saleh, G. O. Darwish, and Y. A.-G. Mahmoud, "Bacteria-powered self-healing concrete: Breakthroughs, challenges, and future prospects," *Journal of Industrial Microbiology and Biotechnology*, vol. 52, Art. no. kuae051, 2025, doi: 10.1093/jimb/kuae051.
- [13] J. Feng, B. Chen, W. Sun, and Y. Wang, "Microbial induced calcium carbonate precipitation study using *Bacillus subtilis* with application to self-healing concrete preparation and characterization," *Construction and Building Materials*, vol. 280, Art. no. 122460, 2021, doi: 10.1016/j.conbuildmat.2021.122460.
- [14] S. Muhammad, H. A. Siddiqui, M. I. Abro, A. Usmani, and M. A. A. Malick, "*Bacillus subtilis* as self-healing agent in cement mortar: The effect of curing time and amount of calcium lactate on strength," *Mehran University Research Journal of Engineering and Technology*, vol. 40, no. 4, pp. 889–897, 2021, doi: 10.22581/muet1982.2104.19.
- [15] S. Muhammad, H. A. Siddiqui, M. I. Abro, A. Usmani, and M. A. A. Malik, "*Bacillus subtilis* as self-healing agent in cement mortar: Combined and the separate effect of bacteria and calcium lactate on self-healing behavior in cement mortar," *Mehran University Research Journal of Engineering and Technology*, vol. 40, no. 2, pp. 426–434, 2021, doi: 10.22581/muet1982.2102.16.

- [16] K. Vijay and M. Murmu, "Effect of calcium lactate on compressive strength and self-healing of cracks in microbial concrete," *Frontiers of Structural and Civil Engineering*, vol. 13, pp. 515–525, 2019, doi: 10.1007/s11709-018-0494-2.
- [17] S. H. Abo Sabah, L. H. Anneza, M. I. Juki, H. Alabduljabbar, N. Othman, A. A. Al-Gheethi, and A. F. Al-Shalif, "The use of calcium lactate to enhance the durability and engineering properties of bio-concrete," *Sustainability*, vol. 13, no. 16, Art. no. 9269, 2021, doi: 10.3390/su13169269.
- [18] V. Wiktor and H. M. Jonkers, "Quantification of crack-healing in novel bacteria-based self-healing concrete," *Cement and Concrete Composites*, vol. 33, no. 7, pp. 763–770, 2011, doi: 10.1016/j.cemconcomp.2011.03.012.
- [19] S. K. Chaerun, R. Syarif, and R. K. Wattimena, "Bacteria incorporated with calcium lactate pentahydrate to improve the mortar properties and self-healing occurrence," *Scientific Reports*, vol. 10, Art. no. 17873, 2020, doi: 10.1038/s41598-020-74127-4.
- [20] S. S. E. Ahmad, M. A. Elshazly, A. A. Elakhras, A. A. Elshami, M. A. R. Elmahdy, and A. Aboubakr, "Thermal dependency of microbial self-healing of concrete: Comparing the effectiveness of *Bacillus* strains at different temperatures, from room temperature to below zero," *Case Studies in Construction Materials*, vol. 23, Art. no. e05182, 2025, doi: 10.1016/j.cscm.2025.e05182.
- [21] Y. Wang, Y. Wang, K. Soga, J. T. DeJong, and A. J. Kabla, "Microscale investigations of temperature-dependent microbially induced carbonate precipitation in the temperature range 4–50°C," *Acta Geotechnica*, vol. 18, pp. 2239–2261, 2023, doi: 10.1007/s11440-022-01664-9.
- [22] M.-T. Nguyen, C. A. Fernandez, M. M. Haider, K.-H. Chu, G. Jian, S. Nassiri, D. Zhang, R. Rousseau, and V.-A. Glezakou, "Toward self-healing concrete infrastructure: Review of experiments and simulations across scales," *Chemical Reviews*, vol. 123, no. 18, pp. 10838–10876, 2023, doi: 10.1021/acs.chemrev.2c00709.
- [23] F. Almohammed, P. Sihag, S. S. Sammen, K. A. Ostrowski, K. Singh, C. V. S. R. Prasad, and P. Zajdel, "Assessment of soft computing techniques for the prediction of compressive strength of bacterial concrete," *Materials*, vol. 15, no. 2, Art. no. 489, 2022, doi: 10.3390/ma15020489.
- [24] K. Vijay and M. Murmu, "Application of artificial neural networks for prediction of microbial concrete compressive strength," *Journal of Building Pathology and Rehabilitation*, vol. 7, Art. no. 1, 2022, doi: 10.1007/s41024-021-00140-3.
- [25] M. K. Elshaarawy, M. M. Alsaadawi, and A. K. Hamed, "Machine learning and interactive GUI for concrete compressive strength prediction," *Scientific Reports*, vol. 14, Art. no. 16694, 2024, doi: 10.1038/s41598-024-66957-3.
- [26] A. T. C. Tran, M. A. Nguyen, and D. K. Tran, "An explainable machine learning model to predict and elucidate the compressive behavior of high-performance concrete," *Results in Engineering*, vol. 11, Art. no. 100245, 2021, doi: 10.1016/j.rineng.2021.100245.
- [27] J. Zhang, Y. Huang, and H. Wang, "A hybrid strategy of AutoML and SHAP for automated and explainable concrete strength prediction," *Case Studies in Construction Materials*, vol. 19, Art. no. e02405, 2023, doi: 10.1016/j.cscm.2023.e02405.
- [28] Y. Aydın, C. Cakiroglu, G. Bekdaş, and Z. W. Geem, "Explainable ensemble learning and multilayer perceptron modeling for compressive strength prediction of ultra-high-performance concrete," *Biomimetics*, vol. 9, no. 9, Art. no. 544, 2024, doi: 10.3390/biomimetics9090544.
- [29] K. S. Sri, R. R. Nayaka, and M. V. N. Siva Kumar, "Mechanical properties of sustainable self-healing concrete and its performance evaluation using ANN and ANFIS models," *Journal of Building Pathology and Rehabilitation*, vol. 8, Art. no. 99, 2023, doi: 10.1007/s41024-023-00345-8.
- [30] M. Meghashree, N. Urs, N. K. Geetha, Tahera, and E. K. Reshma, "Advancing sustainable concrete with bacterial self-healing technology and Kuhn-Tucker condition," *Scientific Reports*, vol. 15, Art. no. 33736, 2025, doi: 10.1038/s41598-025-96971-y.

- [31] K. Cao, D. Liu, K. H. Kong, W. Zhang, Y. Tang, and Y. Jian, "Predicting self-healing efficiency in recycled aggregate concrete using optimized machine learning models," *Scientific Reports*, vol. 15, Art. no. 36831, 2025, doi: 10.1038/s41598-025-12365-0.
- [32] N. De Belie et al., "A review of self-healing concrete for damage management of structures," *Advanced Materials Interfaces*, vol. 5, no. 17, Art. no. 1800074, 2018, doi: 10.1002/admi.201800074.
- [33] V. C. Li and E. Herbert, "Robust self-healing concrete for sustainable infrastructure," *Journal of Advanced Concrete Technology*, vol. 10, no. 6, pp. 207–218, 2012, doi: 10.3151/jact.10.207.
- [34] L. Ferrara, "Self-healing cement-based materials: An asset for sustainable construction industry," *IOP Conference Series: Materials Science and Engineering*, vol. 442, Art. no. 012007, 2018, doi: 10.1088/1757-899X/442/1/012007.
- [35] D. Homma, H. Mihashi, and T. Nishiwaki, "Self-healing capability of fibre reinforced cementitious composites," *Journal of Advanced Concrete Technology*, vol. 7, no. 2, pp. 217–228, 2009, doi: 10.3151/jact.7.217.
- [36] N. K. Dhami, M. S. Reddy, and A. Mukherjee, "Biomineralization of calcium carbonates and their engineered applications: A review," *Frontiers in Microbiology*, vol. 4, Art. no. 314, 2013, doi: 10.3389/fmicb.2013.00314.
- [37] M. J. Castro-Alonso, L. E. Montañez-Hernandez, M. A. Sanchez-Muñoz, M. R. Macias Franco, R. Narayanasamy, and N. Balagurusamy, "Microbially induced calcium carbonate precipitation and its potential in bioconcrete: Microbiological and molecular concepts," *Frontiers in Materials*, vol. 6, Art. no. 126, 2019, doi: 10.3389/fmats.2019.00126.
- [38] P. Anbu, C.-H. Kang, Y.-J. Shin, and J.-S. So, "Formations of calcium carbonate minerals by bacteria and its multiple applications," *SpringerPlus*, vol. 5, Art. no. 250, 2016, doi: 10.1186/s40064-016-1869-2.
- [39] M. Seifan, A. K. Samani, and A. Berenjian, "Induced calcium carbonate precipitation using *Bacillus* species," *Applied Microbiology and Biotechnology*, vol. 100, pp. 9895–9906, 2016, doi: 10.1007/s00253-016-7701-7.
- [40] T. H. Nguyen, E. Ghorbel, H. Fares, and A. Cousture, "Bacterial self-healing of concrete and durability assessment," *Cement and Concrete Composites*, vol. 104, Art. no. 103340, 2019, doi: 10.1016/j.cemconcomp.2019.103340.
- [41] A. A. Al-Gheethi et al., "Critical analysis for life cycle assessment of bio-cementitious materials production and sustainable solutions," *Sustainability*, vol. 14, no. 3, Art. no. 1920, 2022, doi: 10.3390/su14031920.
- [42] A. M. Panza Uguzzoni, E. Fregonara, D. G. Ferrando, G. Anglani, P. Antonaci, and J.-M. Tulliani, "Concrete self-healing for sustainable buildings: A focus on the economic evaluation from a life-cycle perspective," *Sustainability*, vol. 15, no. 18, Art. no. 13637, 2023, doi: 10.3390/su151813637.
- [43] M. Bagga et al., "Advancements in bacteria based self-healing concrete and the promise of modelling," *Construction and Building Materials*, vol. 358, Art. no. 129412, 2022, doi: 10.1016/j.conbuildmat.2022.129412.
- [44] A. Ivaškė, V. Gribniak, R. Jakubovskis, and J. Urbonavičius, "Bacterial viability in self-healing concrete: A case study of non-ureolytic *Bacillus* species," *Microorganisms*, vol. 11, no. 10, Art. no. 2402, 2023, doi: 10.3390/microorganisms11102402.
- [45] I. Ahmad, M. Shokouhian, M. Jenkins, and G. L. McLemore, "Quantifying the self-healing efficiency of bioconcrete using *Bacillus subtilis* immobilized in polymer-coated lightweight expanded clay aggregates," *Buildings*, vol. 14, no. 12, Art. no. 3916, 2024, doi: 10.3390/buildings14123916.
- [46] I. Ahmad, M. Shokouhian, D. Owolabi, M. Jenkins, and G. L. McLemore, "Assessment of biogenic healing capability, mechanical properties, and freeze–thaw durability of bacterial-based concrete using *Bacillus subtilis*, *Bacillus sphaericus*, and *Bacillus megaterium*," *Buildings*, vol. 15, no. 6, Art. no. 943, 2025, doi: 10.3390/buildings15060943.

- [47] M. S. Rahman, S. Rahman, M. S. Chowdhury, M. R. R. Shuvo, and A. Rahman, "Performance analysis of *Bacillus subtilis* implied self-healing cement mortar at local laboratory environment," *Bulletin of the National Research Centre*, vol. 49, Art. no. 68, 2025, doi: 10.1186/s42269-025-01365-9.
- [48] S. S. Lucas, C. Moxham, E. Tziviloglou, and H. Jonkers, "Study of self-healing properties in concrete with bacteria encapsulated in expanded clay," *Science and Technology of Materials*, vol. 30, no. 2, pp. 93–98, 2018, doi: 10.1016/j.stmat.2018.11.006.
- [49] S. A. L. de Koster, R. M. Mors, H. W. Nugteren, H. M. Jonkers, G. M. H. Meesters, and J. R. van Ommen, "Geopolymer coating of bacteria-containing granules for use in self-healing concrete," *Procedia Engineering*, vol. 102, pp. 475–484, 2015, doi: 10.1016/j.proeng.2015.01.193.
- [50] C. Litina et al., "Evaluation of methodologies for assessing self-healing performance of concrete with mineral expansive agents: An interlaboratory study," *Materials*, vol. 14, no. 8, Art. no. 2024, 2021, doi: 10.3390/ma14082024.
- [51] T. Van Mullem, E. Gruyaert, R. Caspeelee, and N. De Belie, "First large scale application with self-healing concrete in Belgium: Analysis of the laboratory control tests," *Materials*, vol. 13, no. 4, Art. no. 997, 2020, doi: 10.3390/ma13040997.
- [52] R. M. Mors and H. M. Jonkers, "Bacteria-based self-healing concrete: Evaluation of full scale demonstrator projects," *RILEM Technical Letters*, vol. 4, pp. 138–144, 2020, doi: 10.21809/rilemtechlett.2019.93.
- [53] A. R. Theja, M. S. Reddy, B. B. Jindal, and C. Sashidhar, "Predicting the strength properties of self healing concrete using artificial neural network," *Journal of Soft Computing in Civil Engineering*, vol. 7, no. 1, pp. 56–71, 2023, doi: 10.22115/SCCE.2022.349792.1478.
- [54] K. C. Onyelowe et al., "Modeling the influence of bacteria concentration on the mechanical properties of self-healing concrete for sustainable bio-concrete structures," *Scientific Reports*, vol. 14, Art. no. 8414, 2024, doi: 10.1038/s41598-024-58666-8.
- [55] F. S. de Oliveira and R. Stefani, "Evaluating the use of synthetic data for machine learning prediction of self-healing capacity of concrete," *AI in Civil Engineering*, vol. 4, Art. no. 25, 2025, doi: 10.1007/s43503-025-00074-6.
- [56] A. Vedrtam, K. Kalauni, and M. T. Palou, "Finite element simulation of bacterial self-healing in concrete using microstructural transport and precipitation modeling," *Scientific Reports*, vol. 15, Art. no. 15809, 2025, doi: 10.1038/s41598-025-99844-6.
- [57] T. Milović, V. Bulatović, L. Pezo, M. Dramićanin, A. Tomić, M. Pezo, and O. Šovljanski, "Enhancing compressive strength of cement by indigenous individual and co-culture *Bacillus* bacteria," *Materials*, vol. 17, no. 20, Art. no. 4975, 2024, doi: 10.3390/ma17204975.
- [58] V. Achal, A. Mukherjee, and M. S. Reddy, "Microbial concrete: Way to enhance the durability of building structures," *Journal of Materials in Civil Engineering*, vol. 23, no. 6, pp. 730-734, 2011, doi: 10.1061/(ASCE)MT.1943-5533.0000159.
- [59] HiMedia Laboratories Pvt. Ltd., "Nutrient Broth M002," Technical Data. [Online]. Available: <https://www.himedialabs.com/media/TD/M002.pdf>. Accessed: Jul. 15, 2026.
- [60] ASTM International, *ASTM C109/C109M-23: Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 50 mm [2 in.] Cube Specimens)*. West Conshohocken, PA, USA: ASTM International, 2023, doi: 10.1520/C0109_C0109M-23.
- [61] C. Qian, H. Chen, L. Ren, and M. Luo, "Self-healing of early age cracks in cement-based materials by mineralization of carbonic anhydrase microorganism," *Frontiers in Microbiology*, vol. 6, Art. no. 1225, 2015, doi: 10.3389/fmicb.2015.01225.
- [62] C. A. Schneider, W. S. Rasband, and K. W. Eliceiri, "NIH Image to ImageJ: 25 years of image analysis," *Nature Methods*, vol. 9, no. 7, pp. 671-675, 2012, doi: 10.1038/nmeth.2089.

- [63] M. Luo, C.-X. Qian, and R.-Y. Li, “Factors affecting crack repairing capacity of bacteria-based self-healing concrete,” *Construction and Building Materials*, vol. 87, pp. 1-7, 2015, doi: 10.1016/j.conbuildmat.2015.03.117.
- [64] H. M. Al-Jabali, W. F. Edris, A. D. Almutairi, A. A.-K. A. Al Sayed, and S. Khairy, “Self-healing concrete reinforced with sisal fibers and based on sustainable *Bacillus subtilis* bacteria calcium lactate-fortified,” *Buildings*, vol. 15, no. 24, Art. no. 4495, 2025, doi: 10.3390/buildings15244495.
- [65] A. Vishal, A. Chepuri, and N. Chandana, “Assessment of bacteria-based self-healing concrete through experimental investigations—A sustainable approach,” *Journal of Materials Science: Materials in Engineering*, vol. 20, Art. no. 15, 2025, doi: 10.1186/s40712-025-00215-w.
- [66] A. R. Suleiman, A. Soliman, and M. L. Nehdi, “Effect of environmental exposure on formation of self-healing products in cracked concrete,” in *Proc. CSCE Annual Conf.*, Fredericton, Canada, 2018.